

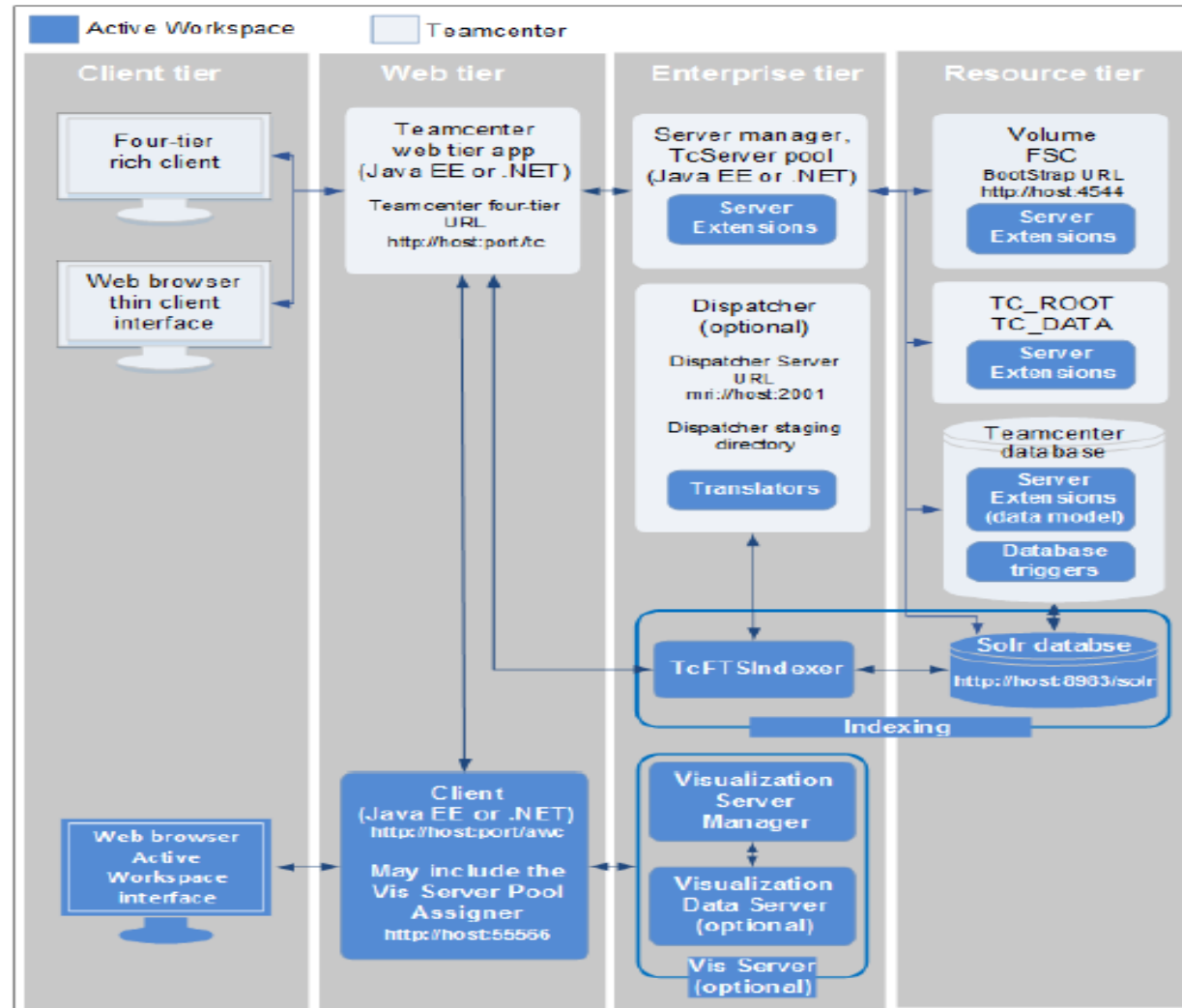
Active Workspace Client Installation/Development Setup

Presented By- Sanjoy

Date – 02/09/2020

1. **Active Workspace components**
2. **Server Extensions Installation**
3. **Indexing**
4. **Visualization Server**
5. **AWC Client Installation**
6. **Development Environment Setup**

1. Active Workspace components



1. Server Extensions
2. Client
3. Indexing
4. Visualization Server

1. Server Extensions

Installing Active Workspace Server Extensions feature adds binaries to TC_ROOT and adds data model changes to the database and TC_DATA. These additions enable Active Workspace functionality. Server Extensions features must be installed on each Teamcenter server that has Teamcenter Foundation installed. This includes corporate servers, volume servers, and server manager hosts

2. Client

Client functionality is provided by a user-facing web application that is built using TEM. Users connect to the Client web application using a web browser. The web application can be either:

- A Java EE-based WAR file (awc.war) deployed on a Java EE web application server such as Jboss/WebSphere/Tomcat.
- A .NET-based web application (awc.zip) deployed on Microsoft IIS.

3. Indexing

The Indexing Server provides the basis for full-text search from the Active Workspace client.

The indexing components are:

1. Indexing Engine

This feature installs the Solr database, which is a full-text search engine that stores indexed Teamcenter data. It is installed by selecting the Indexing Engine feature.

2. Indexer

This feature installs the TcFTSIndexer, which is a four-tier SOA client used to export Teamcenter data for importing into the Solr database. This service manages overall indexing processes. It is installed by selecting the Indexer feature. The TcFTSIndexer is run manually for the initial indexing and can be scheduled to run periodically for subsequent updates.

Standalone Indexing Mode supports indexing both object data and structure data. Indexing operations are run on one machine using a single Java process which connects to the Teamcenter server manager and the Solr search engine.

For custom types, use this mode if the type associated with TcFTSIndexer supports Standalone mode.

4. Visualization Server

The Visualization Server provides dynamic 3D visualization functionality to the Active Workspace client. The Visualization Server is optional. If you plan to use Visualization with Active Workspace, install Visualization Server

The Visualization Server components are:

1. Visualization Server Manager

The Visualization Server Manager streams visualization data to the Active Workspace client. It manages a pool of rendering processes, starting and stopping processes as needed.

2. Visualization Data Server (optional)

The Visualization Data Server improves Visualization performance by caching visualization data close to the Visualization Server Manager. It caches product structure, computes Massive Model Visualization (MMV) spatial hierarchies, and prepopulates JT files in the FCC for faster visualization streaming to the Active Workspace client. You can use the Visualization Server Manager without this component.

A single Visualization Data Server can support one or more Visualization Server Managers. A Visualization Server Manager must be installed on the same host as the Visualization Data Server.

The Visualization Data Server is required for the MMV feature in Active Workspace. Additionally, you need to index structure data for the product configurations that you want to view with the MMV flag.

Active Workspace software components require the following system software and hardware

1. Server Extensions

The corporate server must have a 64-bit operating system and 64-bit JDK.

The Teamcenter four-tier web application (Java EE) must be deployed.

2. Client (Java EE Client)

- The machine on which the WAR file is built must have:

A minimum of 1 GB of free physical RAM (unused by other applications) to avoid paging. Siemens PLM Software recommends that the machine used have a minimum of 2 GB.

- Windows server operating system.

If you are running Teamcenter in a Linux environment, you can deploy the WAR file on a Linux server; however, the WAR file must be built on a Windows server machine.

- 64-bit JDK 1.7 installed and the **JAVA_HOME** system environment variable must be set.

3. Indexing

- The Indexing Engine must be installed on a machine with a 64-bit operating system.
- The Indexing Engine must be installed on a machine with 64-bit JDK or JRE.
- In AIX and Solaris environments, Solr must be installed on a machine with a supported version of Bash.

4. Visualization Server

- **Visualization Server Manager**

- Windows Server 64-bit 2012 R2 or Windows Server 64-bit 2008 R2.

Some graphics cards are not supported on Windows Server 64-bit 2008 R2

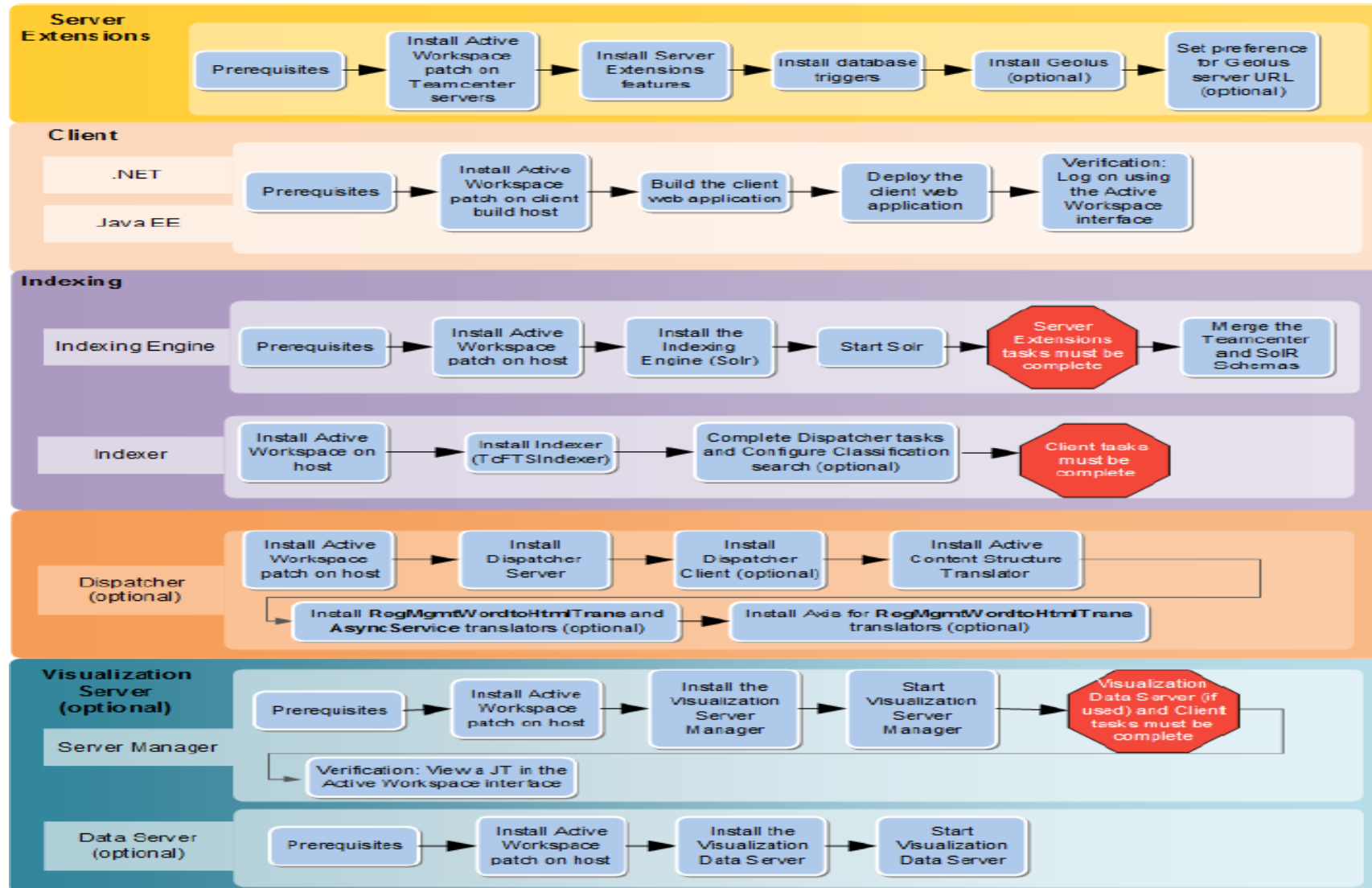
- **Visualization Data Server(Optional)**

- Windows Server 64-bit 2012 R2 or Windows Server 64-bit 2008 R2.

- The machine hosting the Visualization Data Server should have a minimum of 16 GB of RAM but may require more. The amount of RAM needed depends on the number of structures to be indexed and their size.

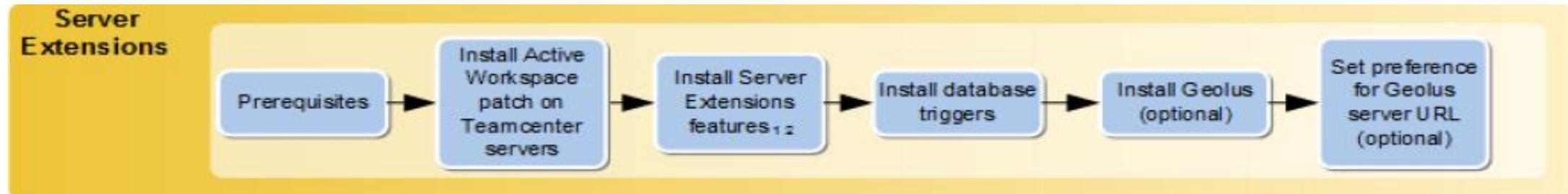
Note:- For versions of system software and hardware certified for running Teamcenter and Active Workspace on your platform, see the **hardware and software certifications** page on GTAC:

Installation overview



Server Extensions Overview

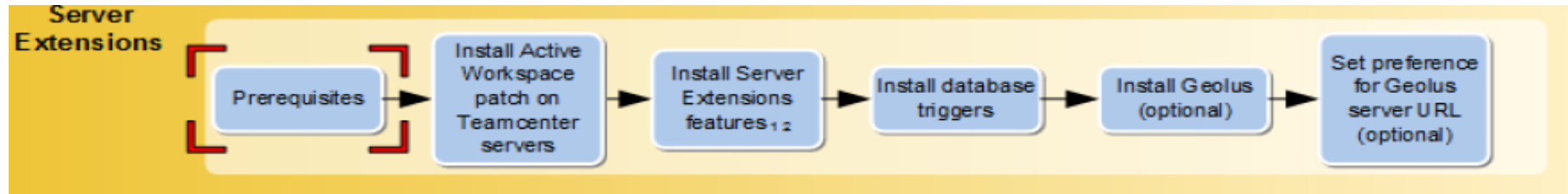
Installing Active Workspace Server Extensions features adds binaries to *TC_ROOT* and adds data model changes to the database and *TC_DATA*. These additions enable Active Workspace functionality.



Values required before beginning the Server Extensions taskflow		
Parameter	Where value is defined	Your value
Indexing Engine user name and password	<i>Install Indexing Engine (Solr)</i>	
Search engine URL (http://host:8983/solr)	<i>Install Indexing Engine (Solr)</i>	
Database system user name and password	Teamcenter database creation	

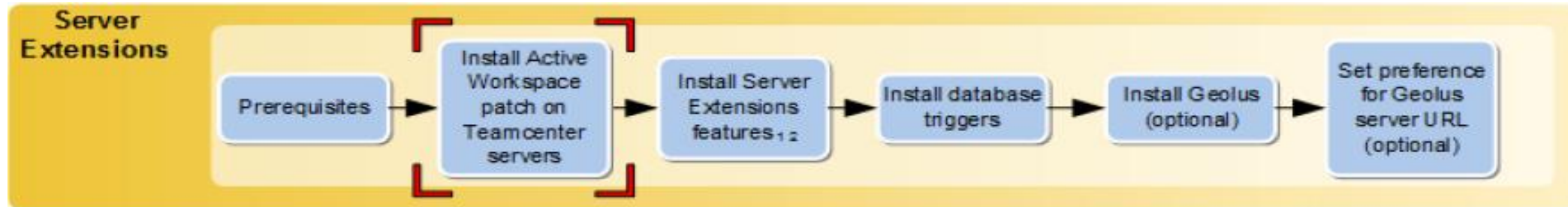
Values to record during the Server Extensions taskflow execution		
Parameter	Where value is defined	Your value
Geolus server URL (only if Shape Search is used)	<i>Install Geolus (optional)</i>	

Server Extensions prerequisites



Values from the Teamcenter four-tier environment	
Parameter	Your value
Teamcenter four-tier URL (http://host:port/tc)	
FSC bootstrap URLs (http://host:4544)	
Database system user login/password	
TC_DATA	
TC_ROOT (for Dispatcher-based indexing)	
SPLM license server host and port (28000@host) (for Dispatcher-based indexing)	

Install the Active Workspace patch on the Teamcenter servers

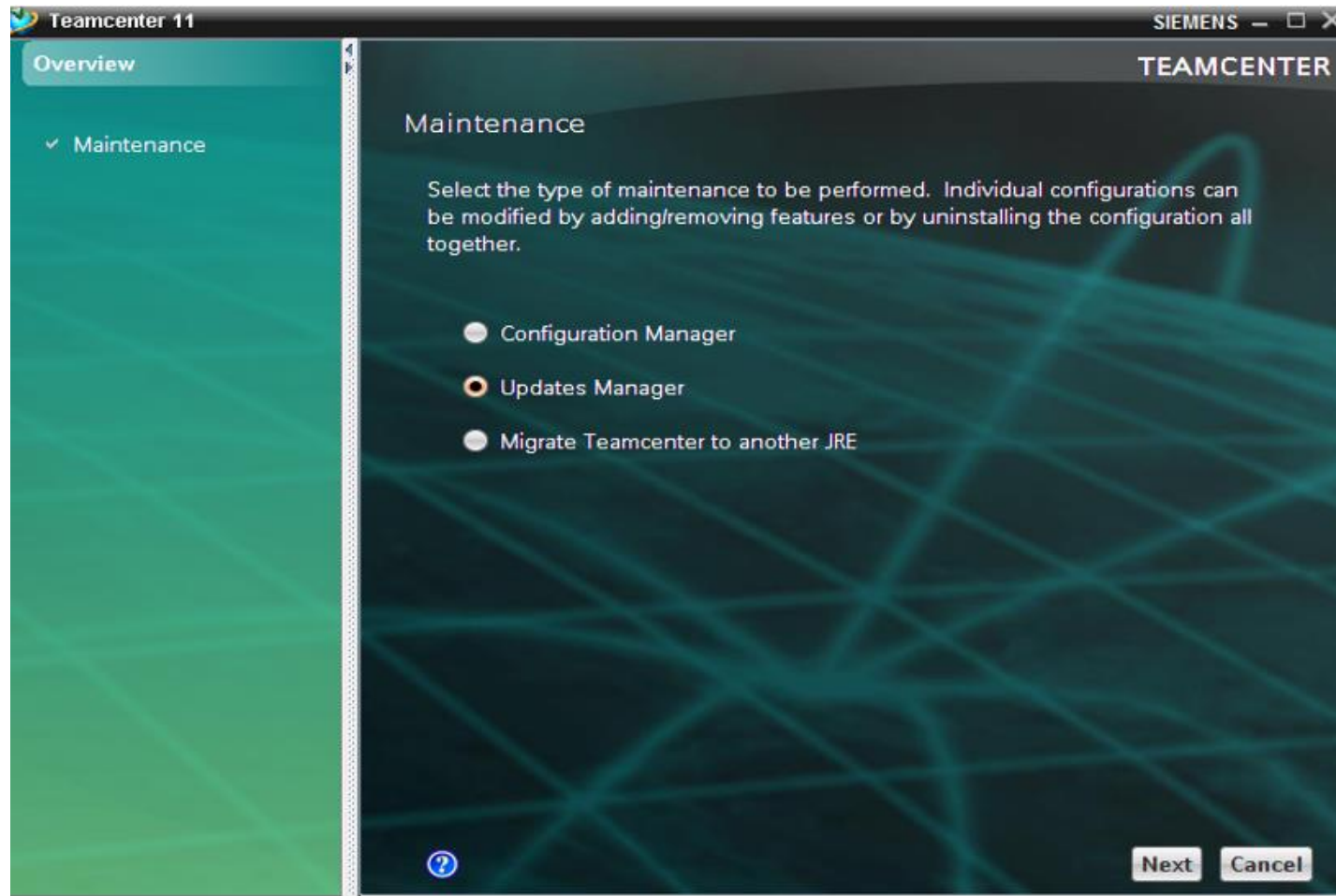


Before you install any Active Workspace features, you must add the Active Workspace patch to all Teamcenter servers that have the Teamcenter Foundation feature installed. This makes Active Workspace features available for installation in TEM.

1. Get the Active workspace software dump (eg. Tc12.0.0.0_ActiveWorkspace4.2.1_Inx64.zip)
2. Unzip the above zip file in some directory
3. Stop all Teamcenter services except FSC services before proceeding with the patch. For example, make sure all server manager services are stopped.
4. Launch TEM for the corporate server.

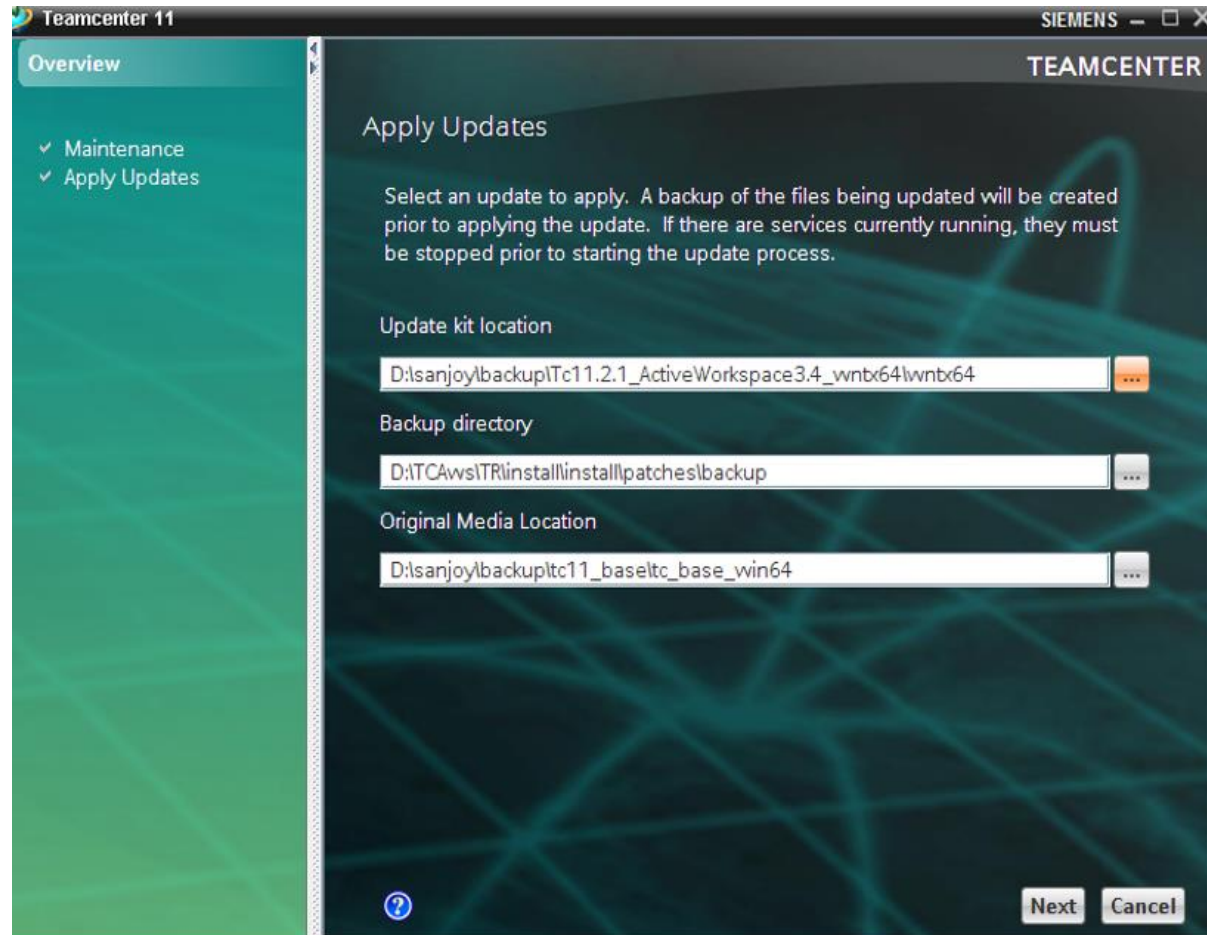
Install the Active Workspace patch on the Teamcenter servers

5. In the Maintenance Panel select Updates Manager



Install the Active Workspace patch on the Teamcenter servers

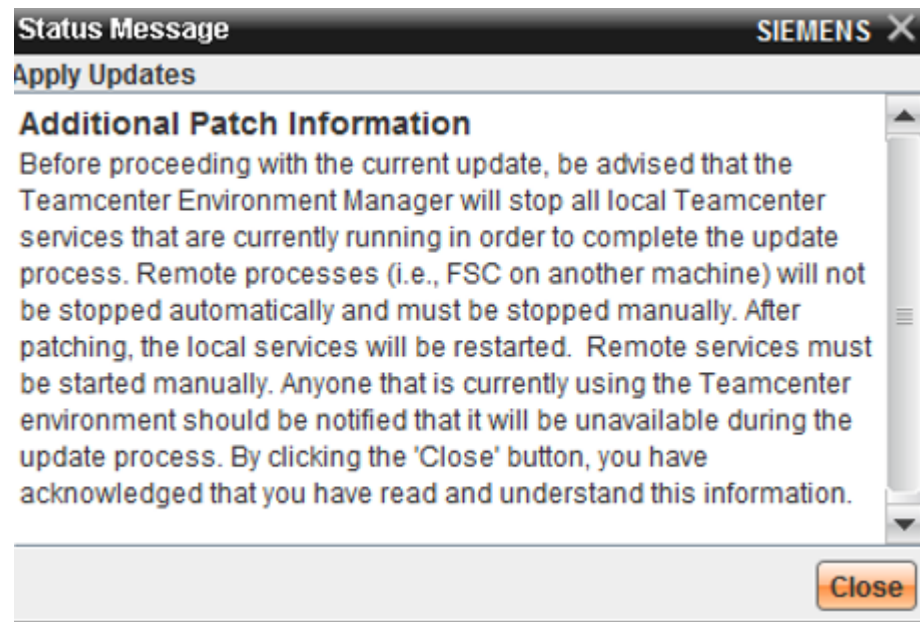
6. Proceed to the **Apply Updates** panel and enter required paths:



- In the **Update kit location** box, enter the path to the expanded Active Workspace software kit.
- In the **Backup directory** box, enter a path in which you want TEM to create backups of files replaced during patching.
- In the **Original Media Location** box, enter the path to the software kit for the major version of Teamcenter

Install the Active Workspace patch on the Teamcenter servers

7. A status message in TEM reminds you that Teamcenter services must be stopped to perform the update. Stop any running Teamcenter services and make sure any open user sessions are closed before you continue. TEM restarts Teamcenter services after the update is completed



Install the Active Workspace patch on the Teamcenter servers

8. Enter OS user and password details

Teamcenter 11

SIEMENS

TEAMCENTER

TCAws11_2_3

Overview

- ✓ Maintenance
- ✓ Apply Updates
- ✓ Optional Configuration
- ✓ OS User
 - Teamcenter Administra
 - Product Excellence
 - Confirmation
 - Executing

Operating System User

The following user will be granted 'Log on as a service' right and all services will run as this user

User: MYTTL\smm914024

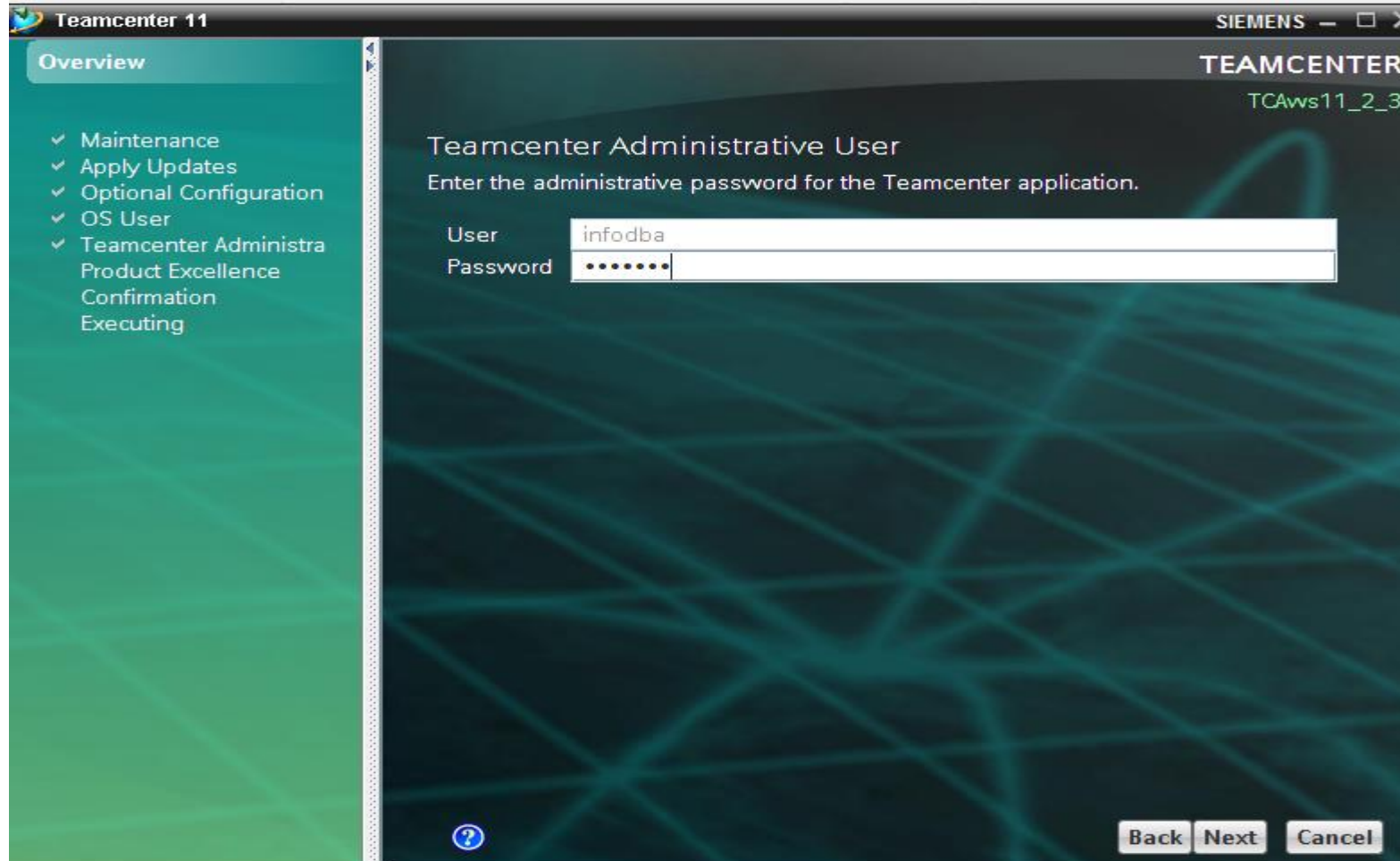
Password:

Confirm:

Back Next Cancel

Install the Active Workspace patch on the Teamcenter servers

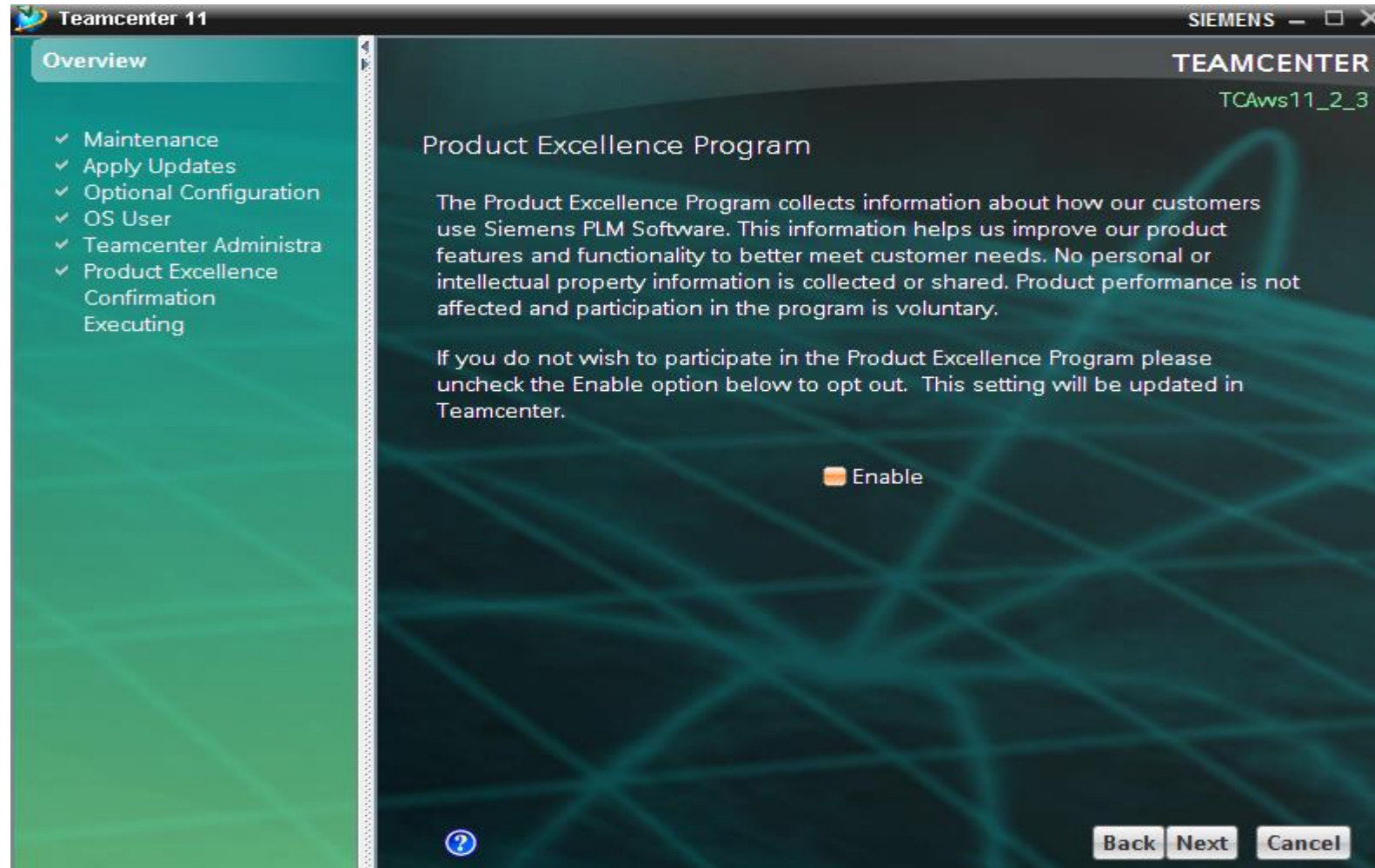
9. Enter Teamcenter Administrative password



The screenshot shows the Teamcenter 11 administrative interface. On the left is a teal sidebar with a menu: 'Overview' (selected), 'Maintenance', 'Apply Updates', 'Optional Configuration', 'OS User', 'Teamcenter Administra', 'Product Excellence', 'Confirmation', and 'Executing'. The main area has a dark teal background with a network pattern. It displays 'TEAMCENTER' and 'TCAws11_2_3' in the top right. The title 'Teamcenter Administrative User' is centered, followed by the instruction 'Enter the administrative password for the Teamcenter application.' Below this are two input fields: 'User' with the text 'infodba' and 'Password' with masked characters '.....'. At the bottom right are three buttons: 'Back', 'Next', and 'Cancel'. A help icon (?) is at the bottom left.

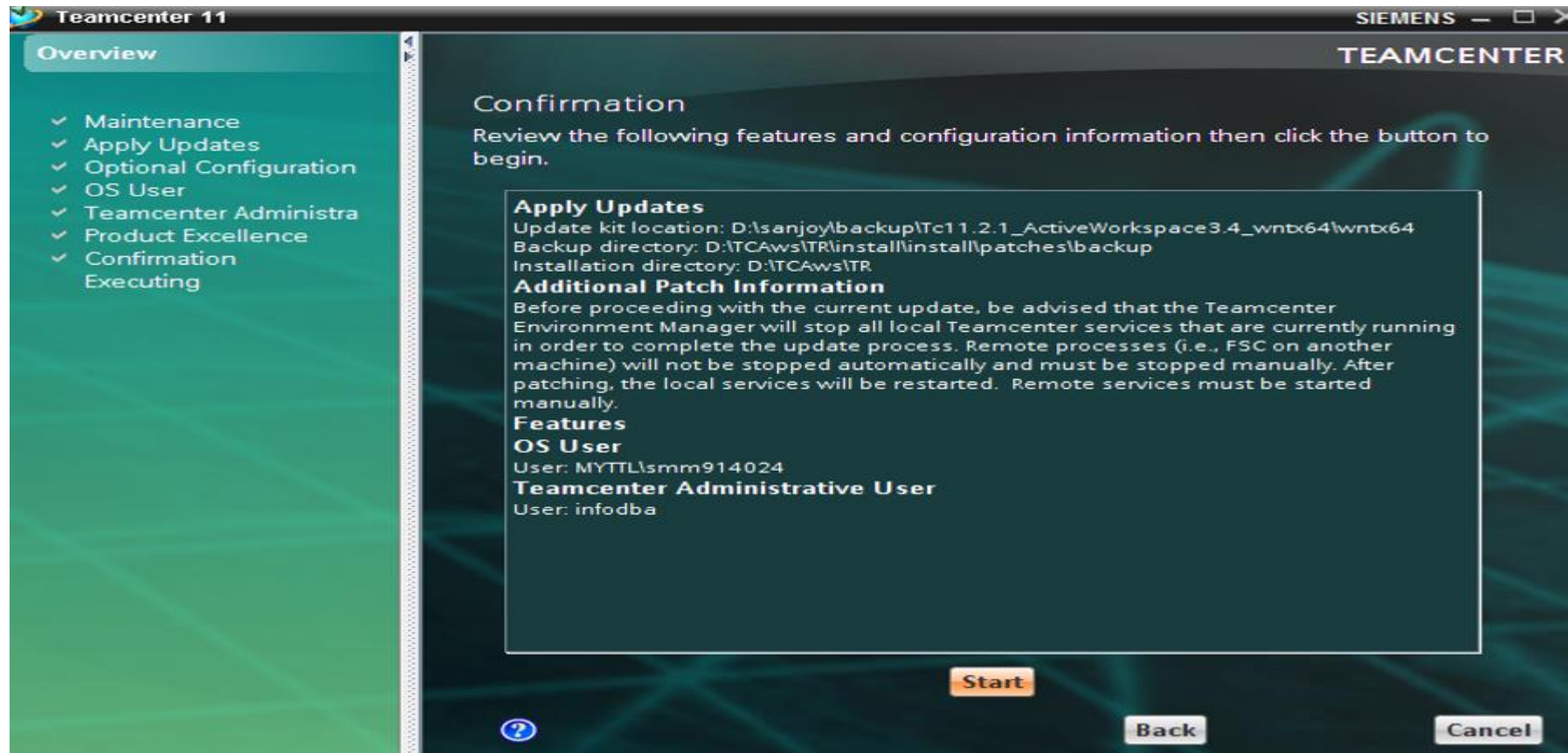
Install the Active Workspace patch on the Teamcenter servers

10. Enter Next.



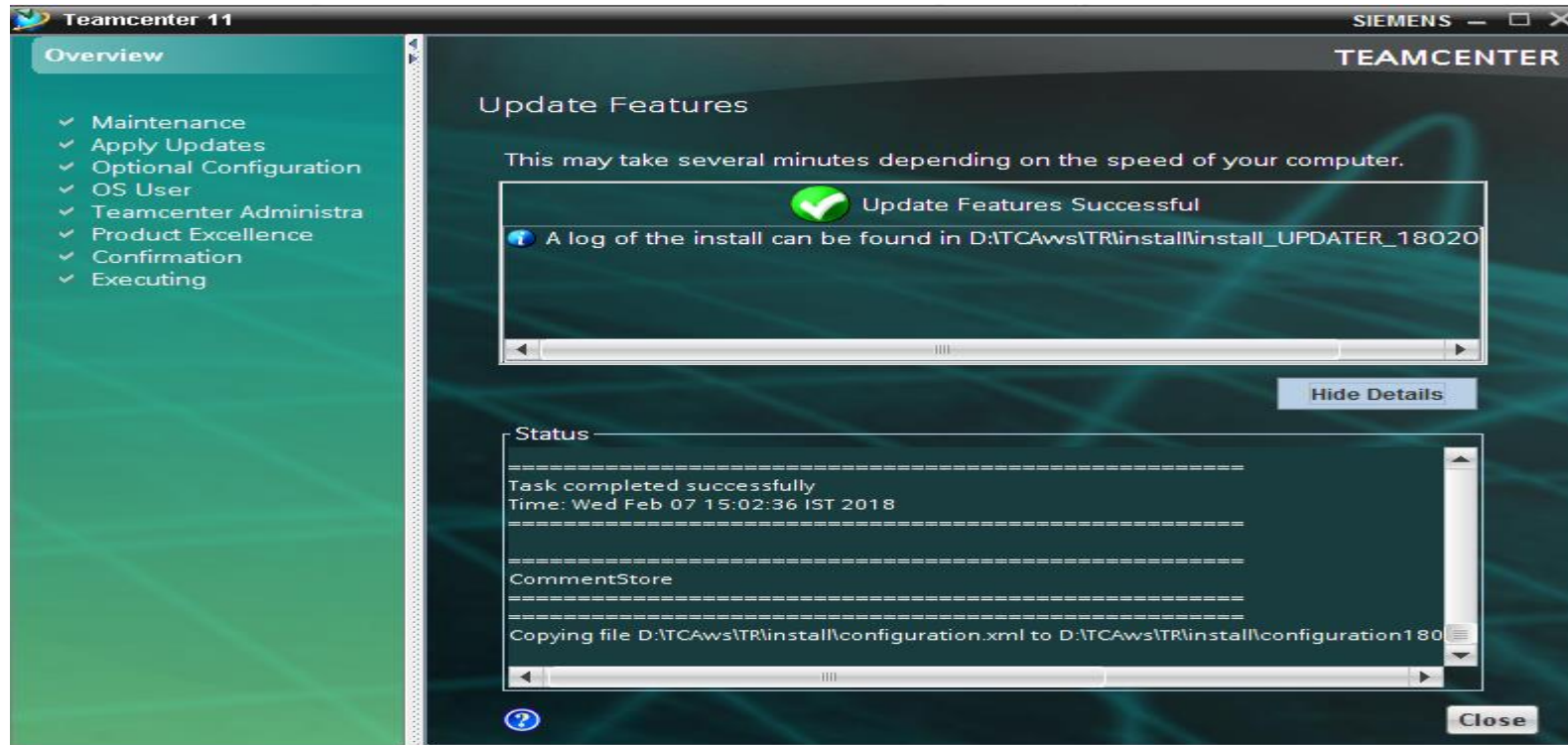
Install the Active Workspace patch on the Teamcenter servers

10. Proceed to the Confirmation panel and click Start to apply the Active Workspace patch



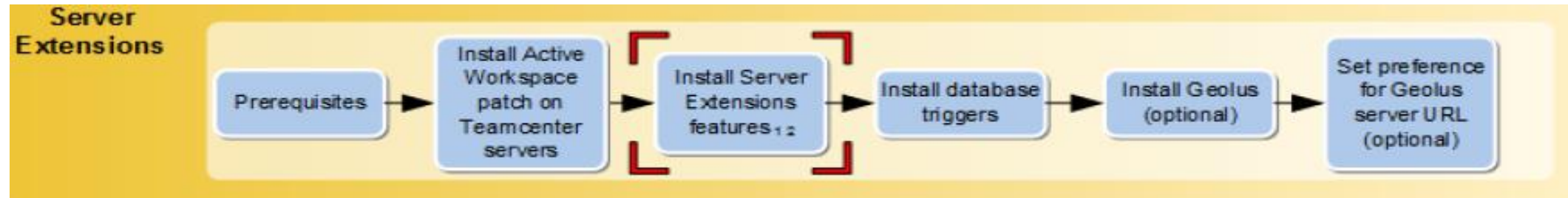
Install the Active Workspace patch on the Teamcenter servers

11. When the update is complete, close TEM



After this In the location of the expanded Active Workspace software kit, browse to the **tc** directory.
Expand the **data.zip** file from this location to **TC_DATA** folder.

Install Server Extensions features

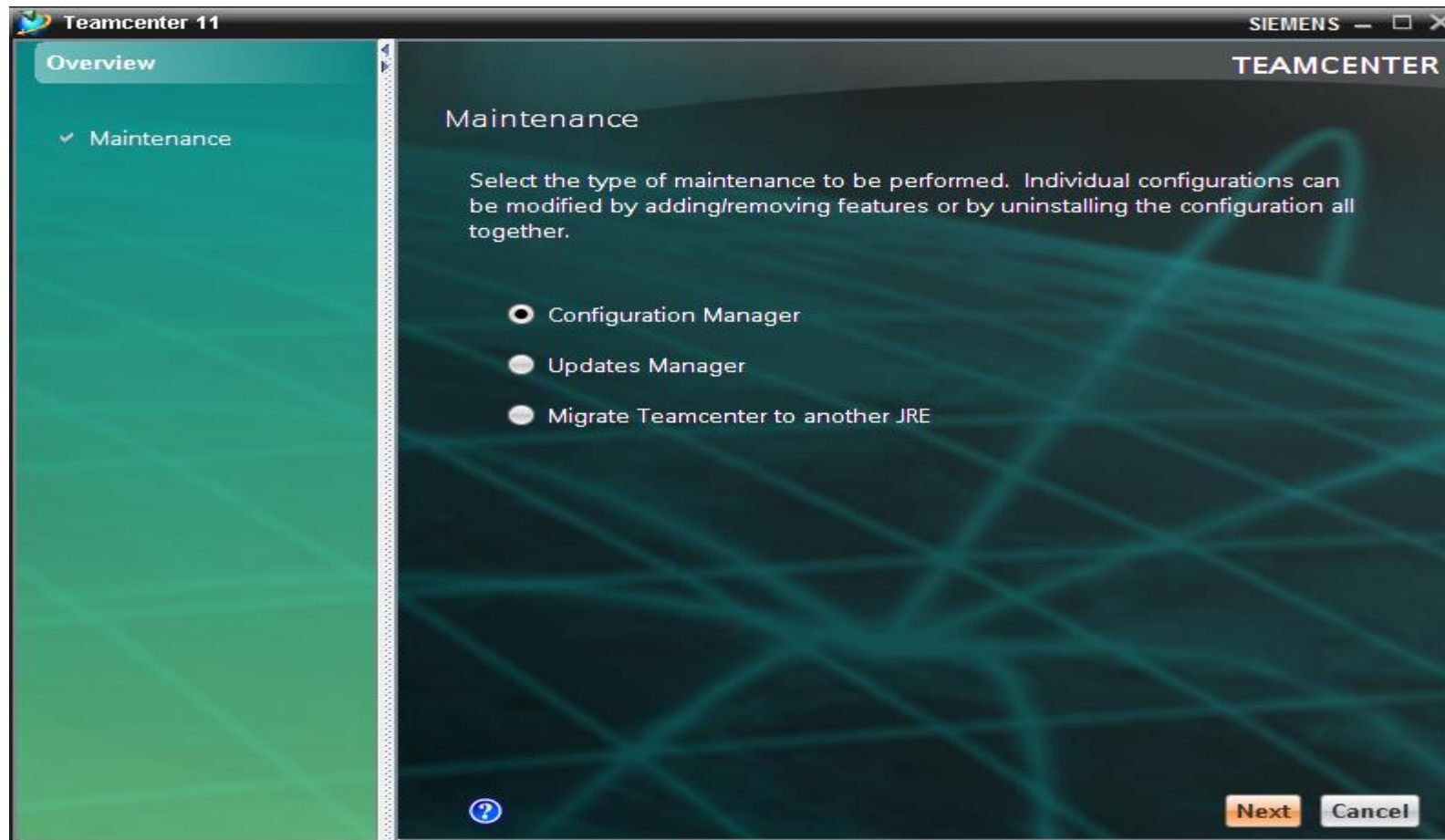


Server Extensions must be installed on every host that has the Teamcenter Foundation feature installed.

1. Stop all Teamcenter services except FSC services before beginning installation.
2. Launch Teamcenter Environment Manager (TEM) for the corporate server to which Server Extensions features are to be added.

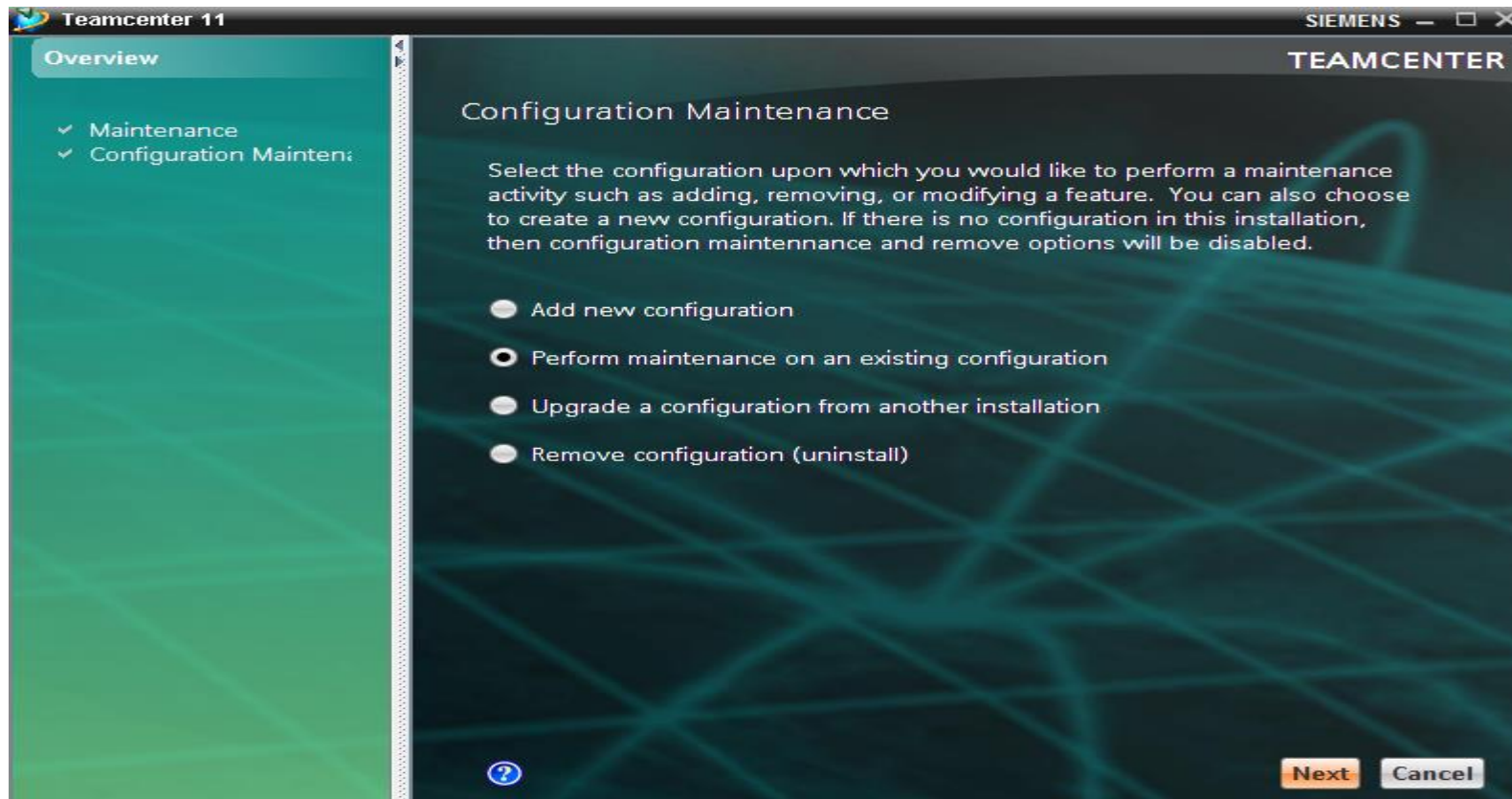
Install Server Extensions features

3. In the Maintenance panel, select Configuration Manager



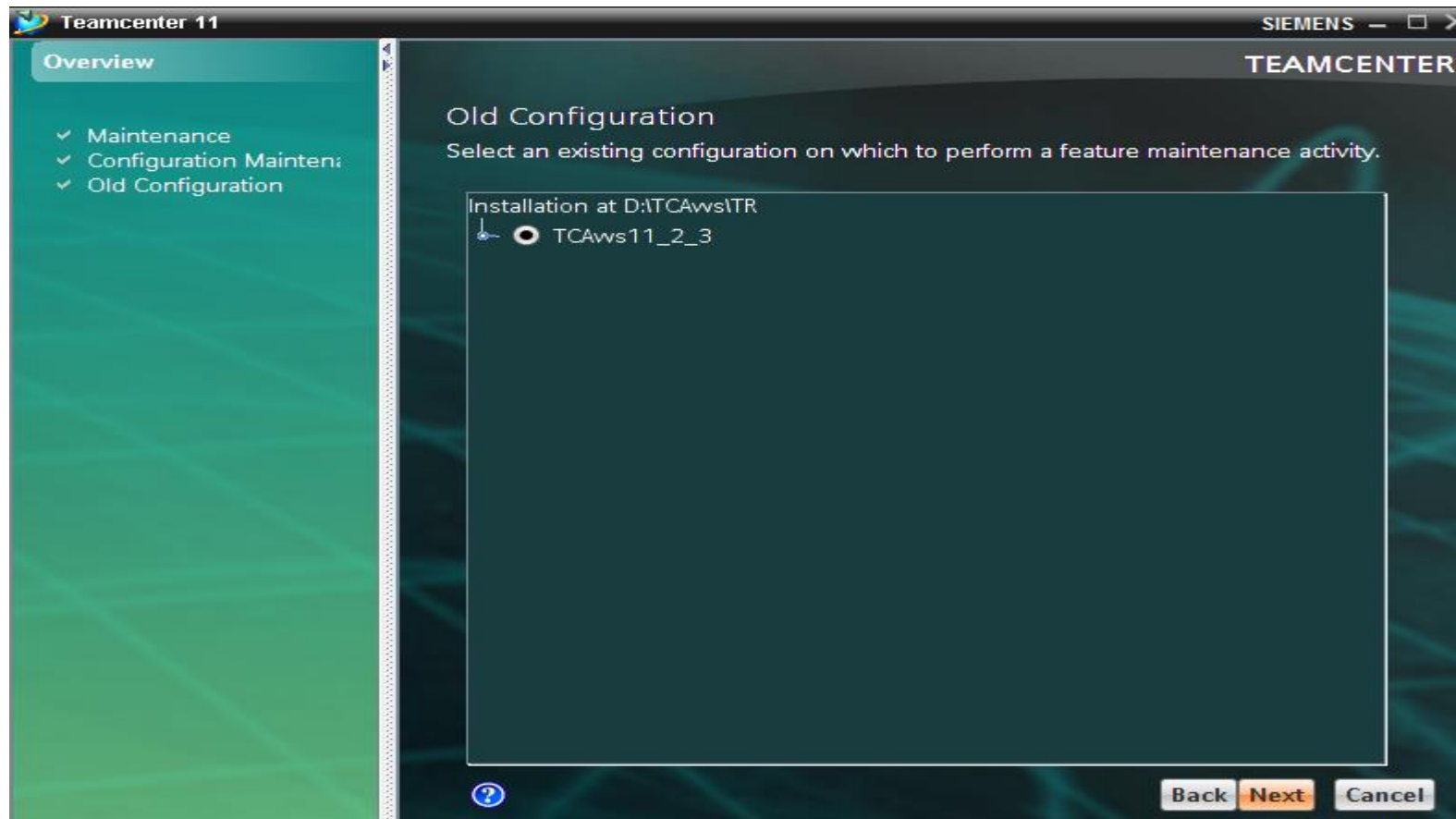
Install Server Extensions features

4. In the **Configuration Maintenance** panel, select **Perform maintenance on an existing configuration**.



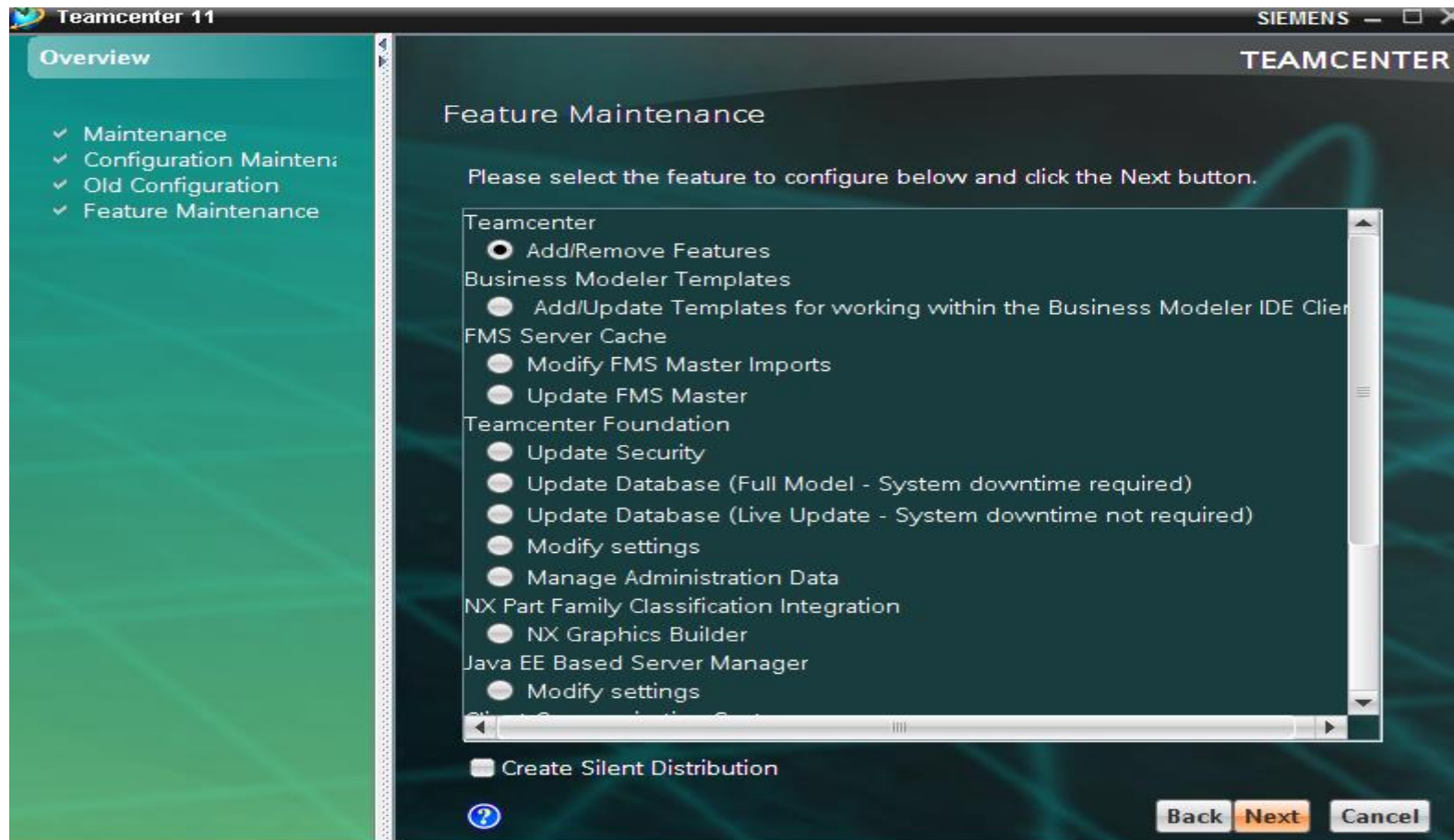
Install Server Extensions features

5. In the **Old Configuration** panel, select the corporate server configuration



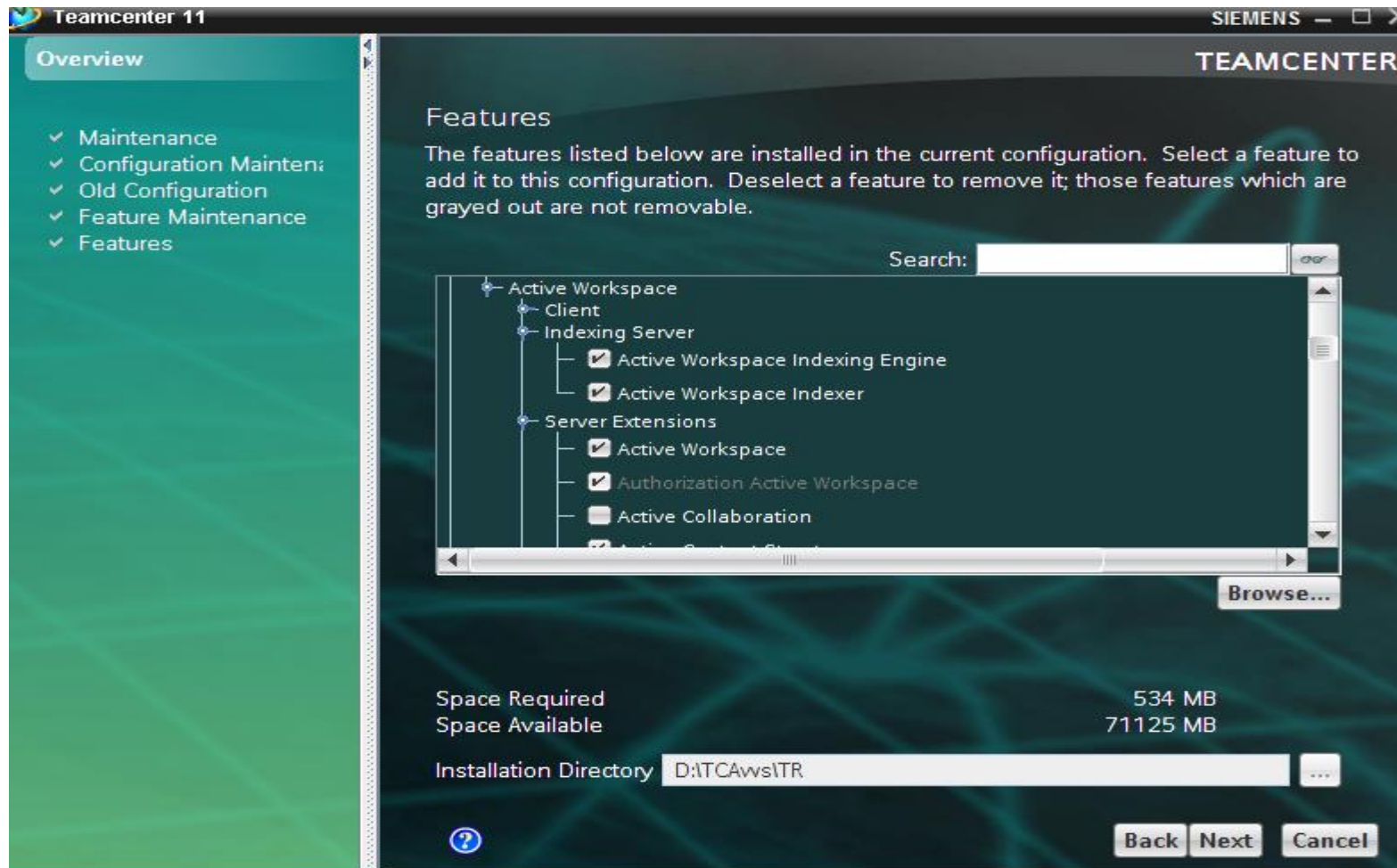
Install Server Extensions features

6. In the **Feature Maintenance** panel, select **Add/Remove Features**.



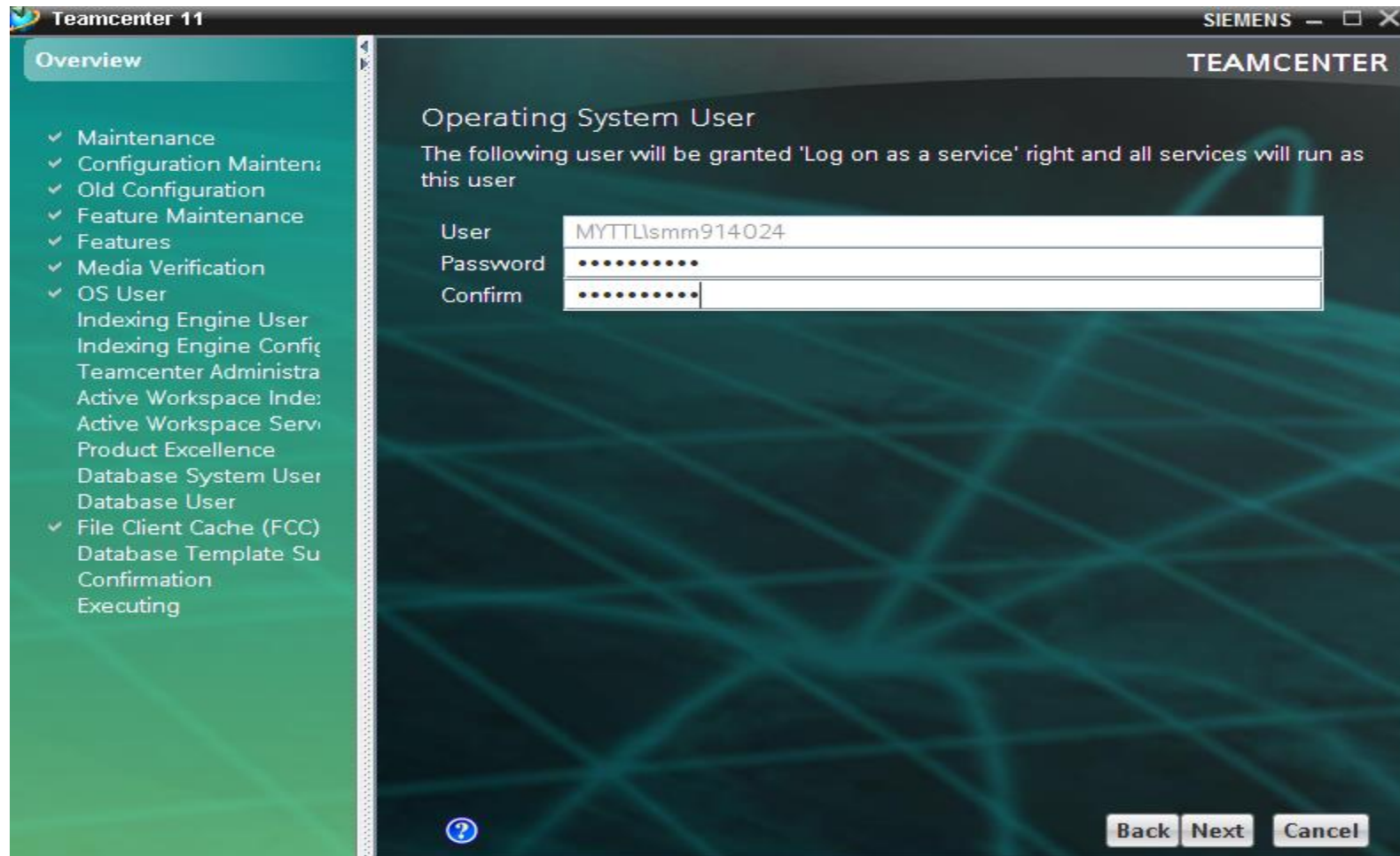
Install Server Extensions features

7. In the **Features** panel, select Active Workspace Server Extensions features:.



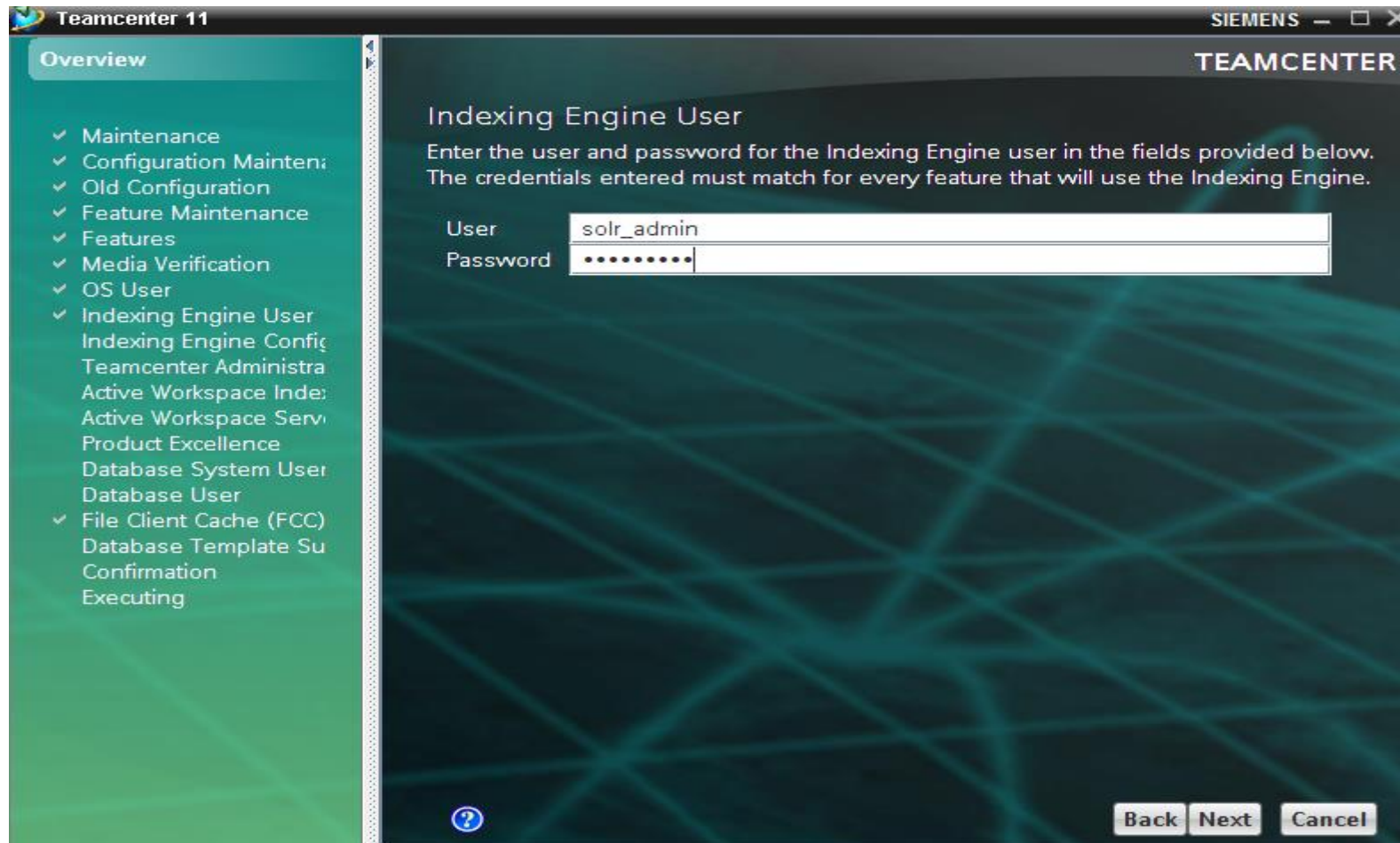
Install Server Extensions features

8. In the **Teamcenter Administrative User** panel, type the user's password:.



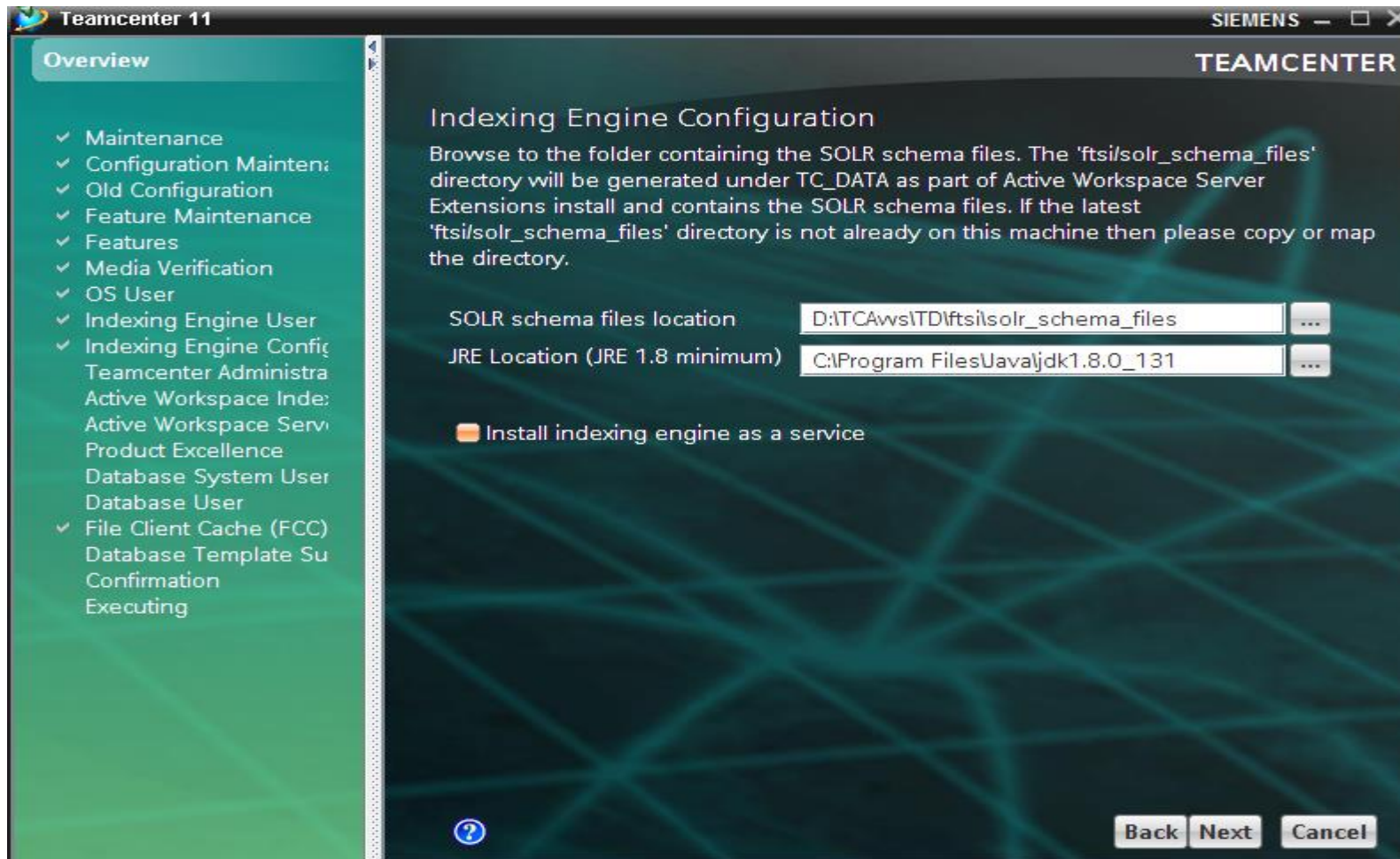
Install Server Extensions features

9. In the **Indexing Engine User** panel, type the Solr administrator's user name and password



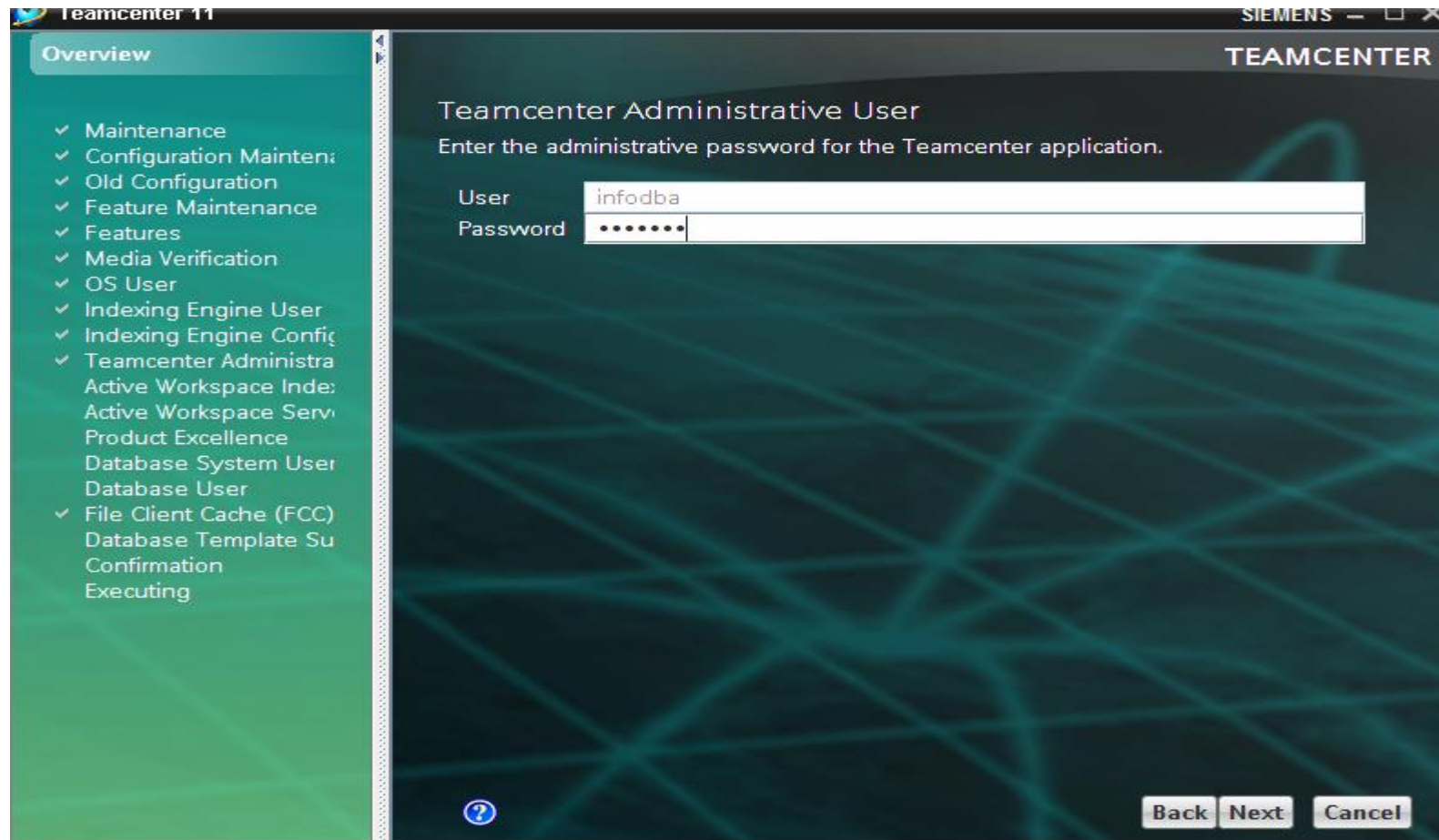
Install Server Extensions features

10. In the **Indexing Engine Configuration** enter the SOLR schema files location and JRE Location



Install Server Extensions features

11. Enter Teamcenter Administrative User password



Install Server Extensions features

12. In the Active Workspace Indexer Settings enter the below details.

Teamcenter 11 SIEMENS

TEAMCENTER

Active Workspace Indexer Settings

Specify Active Workspace Indexer installation settings below. Maximum Teamcenter Connections specifies the maximum number of open connections between the Teamcenter server and the indexer at a given time. It should be less than the number of warmed up Teamcenter servers available in the server manager. . Teamcenter Retry Count specifies the number of tries to connect to the Teamcenter server.

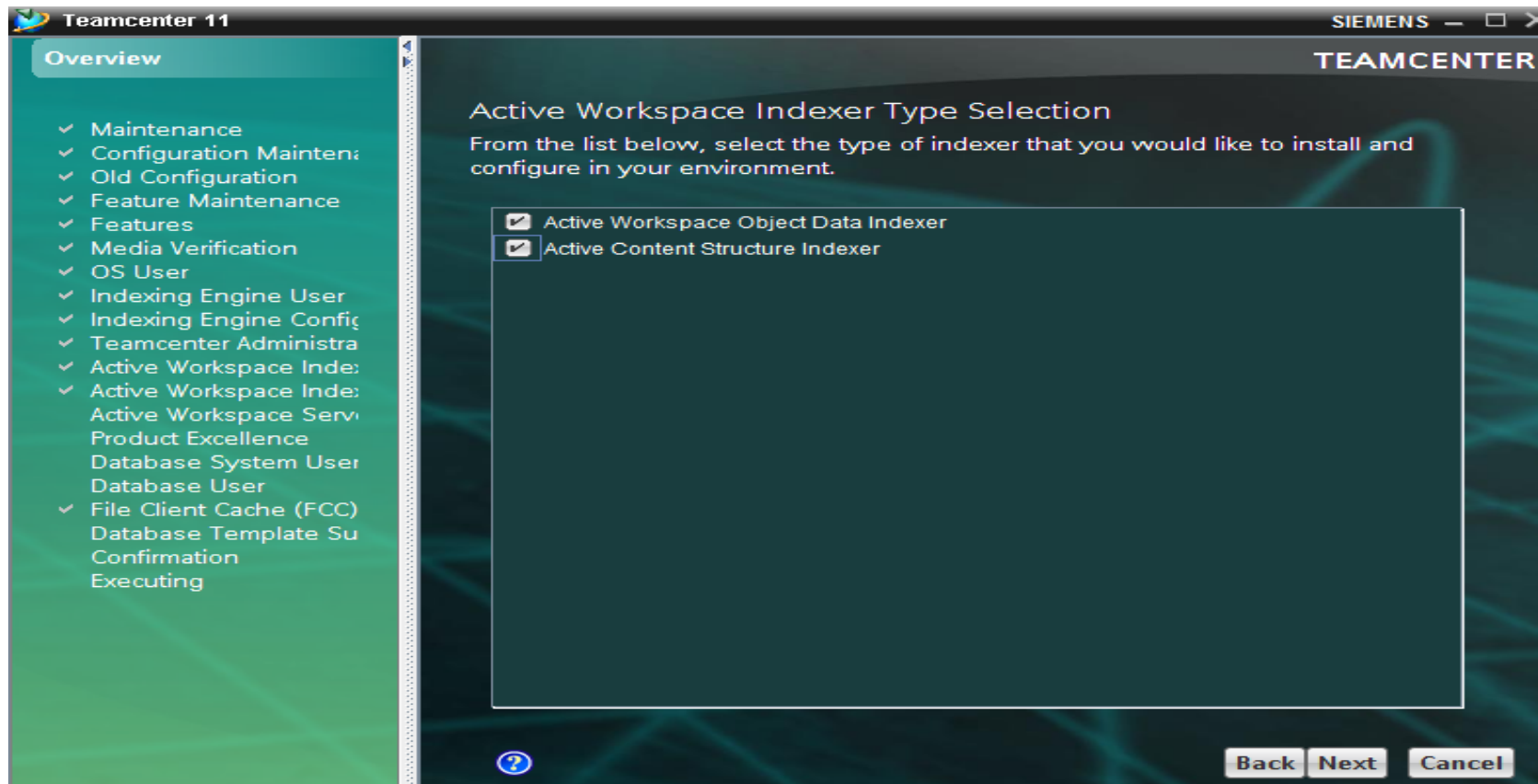
☐ Standalone indexing environment
☐ Dispatcher-based indexing environment

Teamcenter 4-tier URL:
 Maximum Teamcenter Connections:
 Teamcenter Retry Count:
 Dispatcher Server URL:
 Staging Directory: ...

Back Next Cancel

Install Server Extensions features

13. In the Active Workspace Type Selection select the type of indexer you would like to install.



Install Server Extensions features

14. In the Active Workspace Object Data Indexer Settings enter as per your requirements.

The screenshot shows the 'Active Workspace Object Data Indexer Settings' dialog box in Teamcenter 11. The left sidebar lists various maintenance tasks, with 'Active Workspace Object Data Indexer' selected. The main panel contains settings for extraction start and end times.

Active Workspace Object Data Indexer Settings
Configure the settings for extraction start and end times in the fields provided below.

Extraction Time

Start Time

Date: YEAR (2018), MONTH (02), DAY (01)
Time: 09:00
Example: 2015-01-01 Example: 13:15

☒ No End Time

☐ End Time

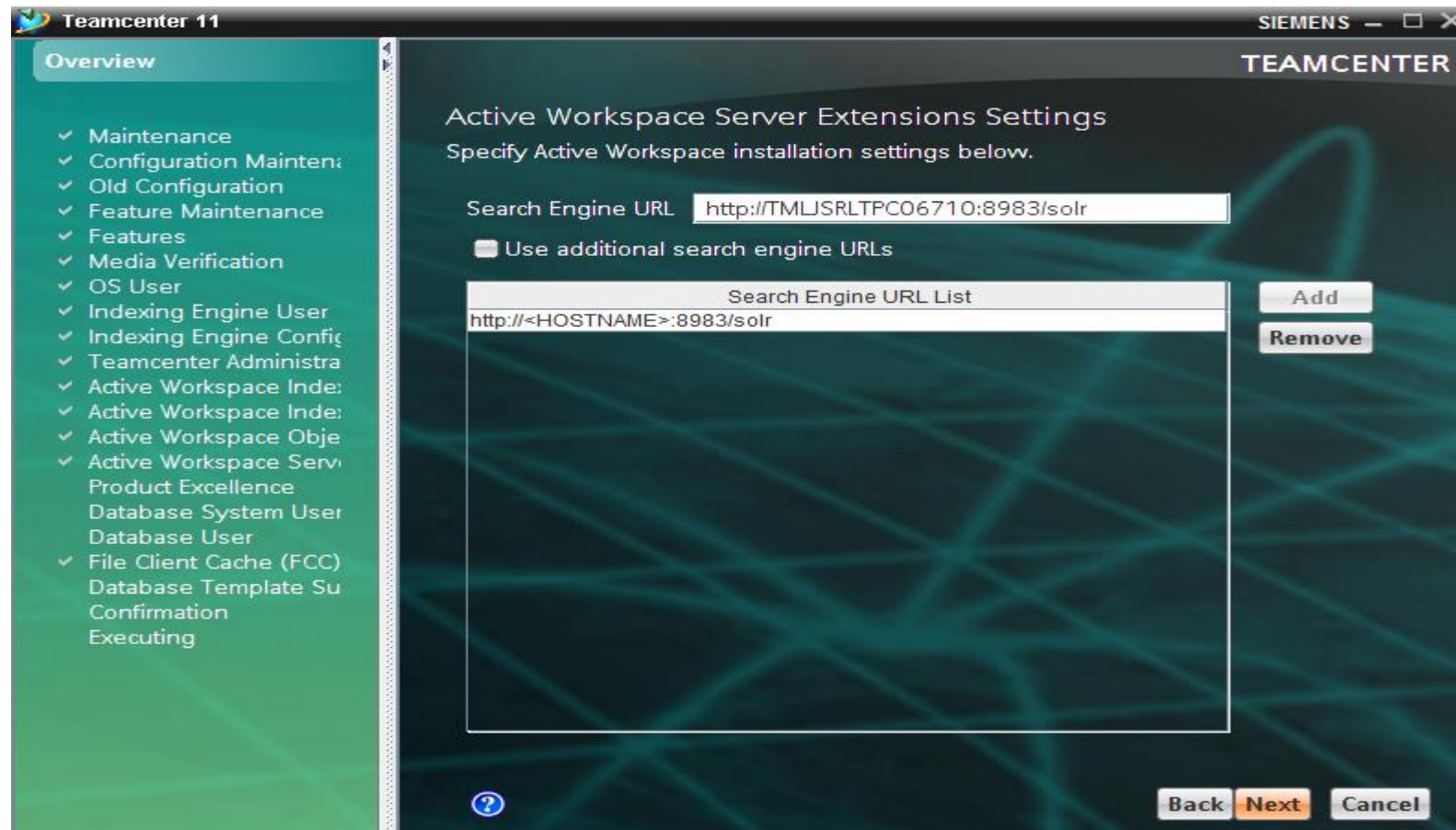
Date: YEAR (2018), MONTH (02), DAY (01)
Time: HH:mm
Example: 2015-01-01 Example: 13:15

If 'No End Time' is specified then indexing will be done for all items that have been created from the specified start time until now.

Buttons: Advanced..., Back, Next, Cancel

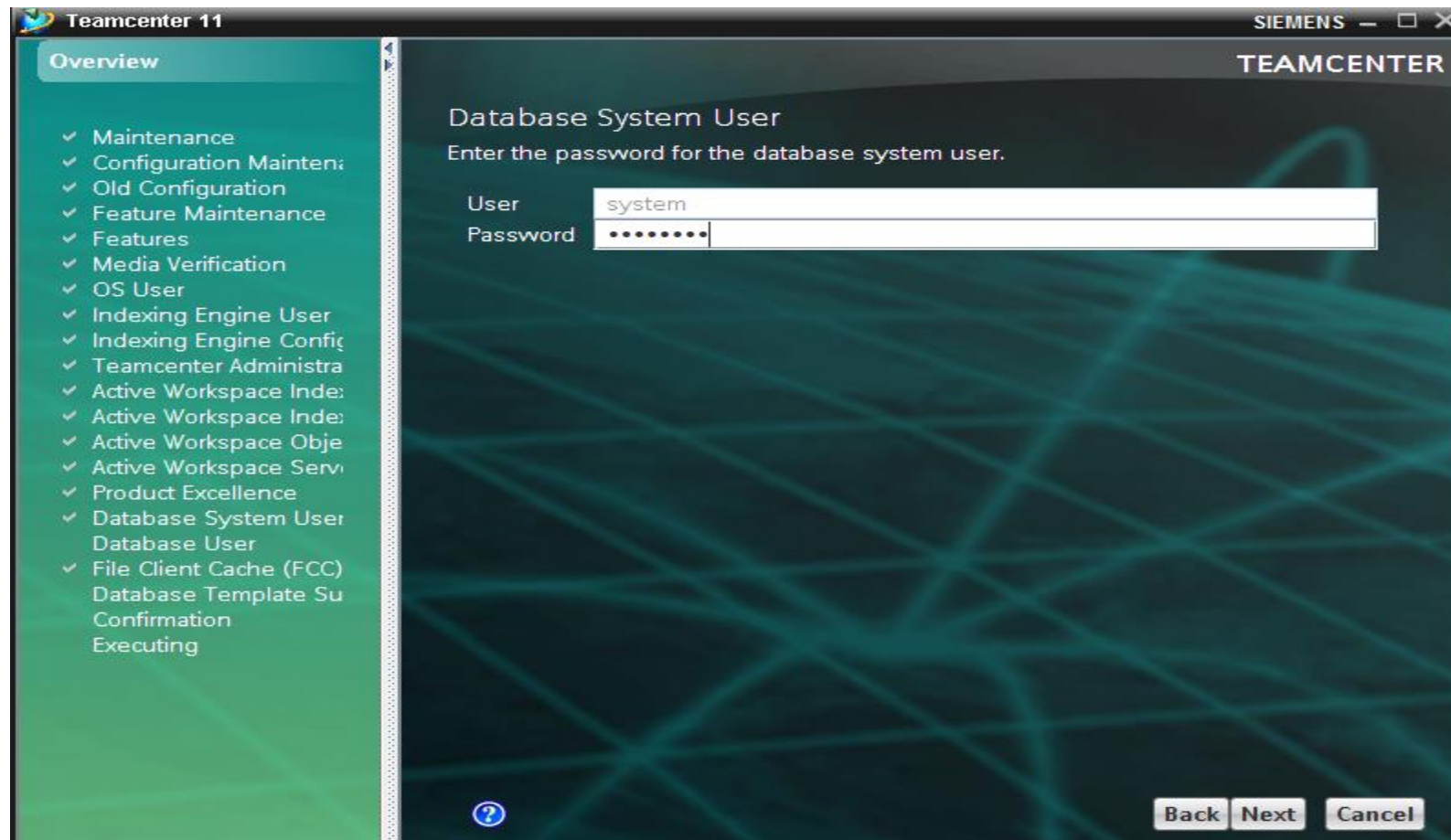
Install Server Extensions features

15. In the **Active Workspace Server Extensions Settings** panel, type the Solr search engine URL.



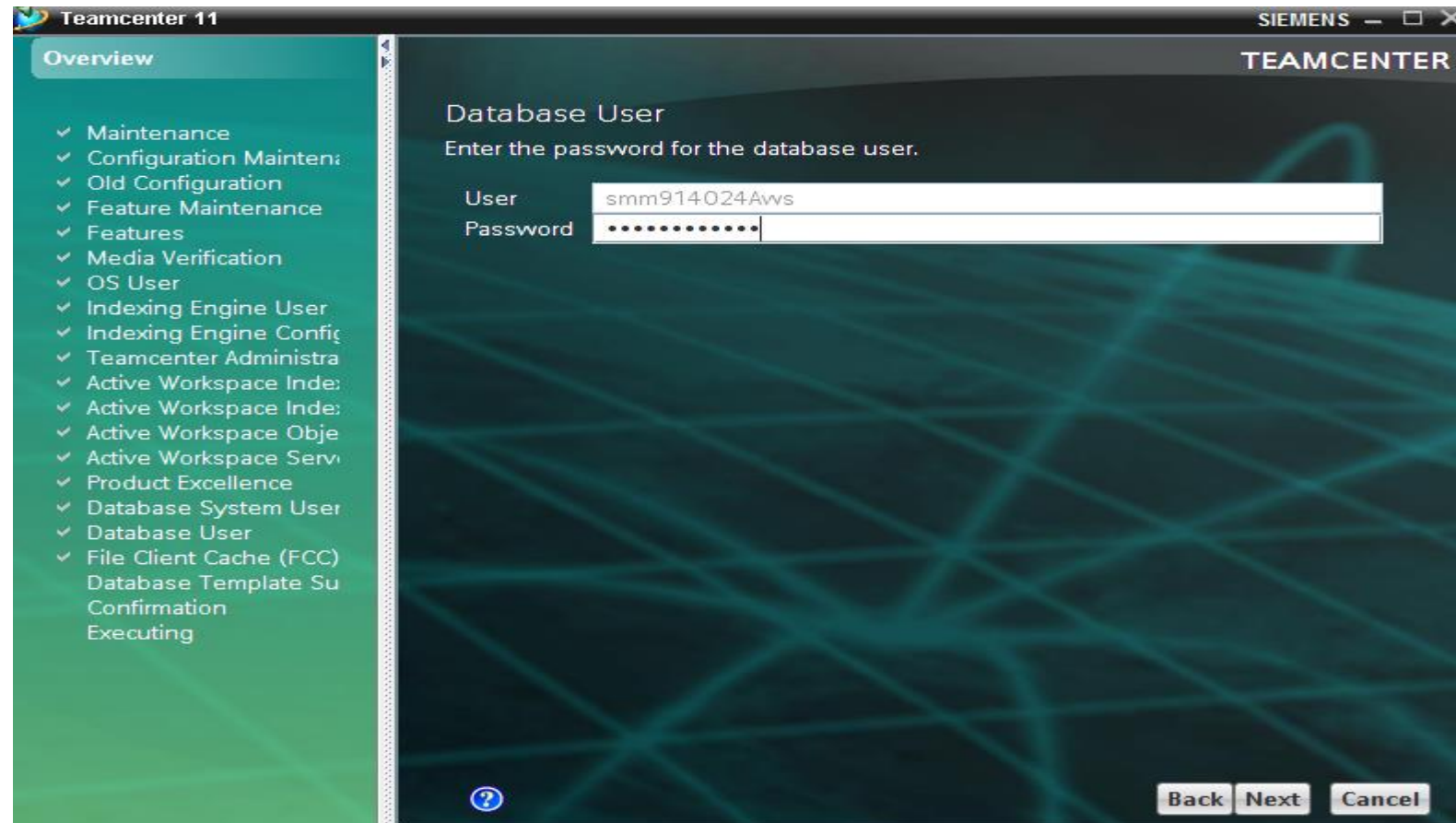
Install Server Extensions features

16. In the **Database System User** panel, type the password for the system database user



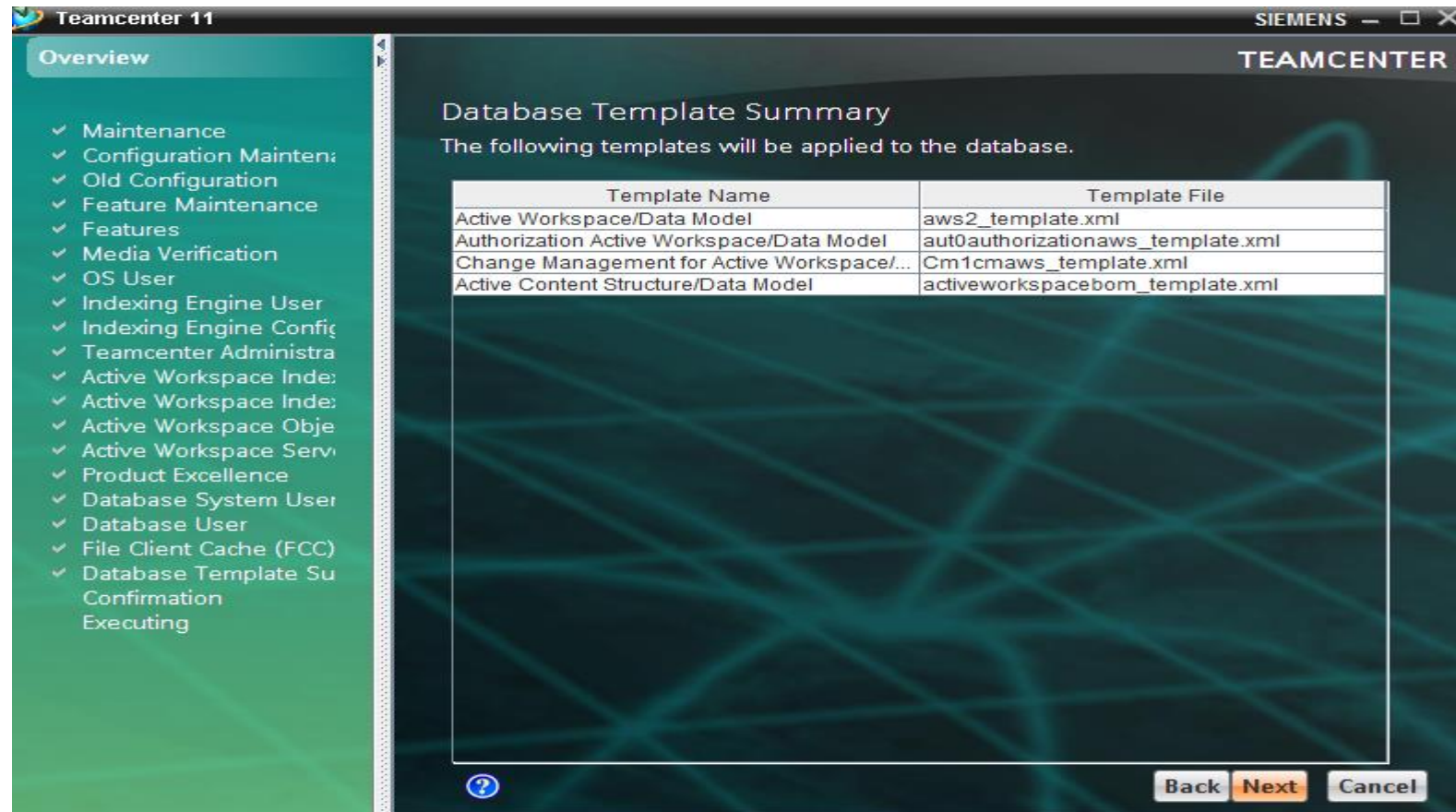
Install Server Extensions features

17. In the **Database User** panel, type the password for the Teamcenter database user



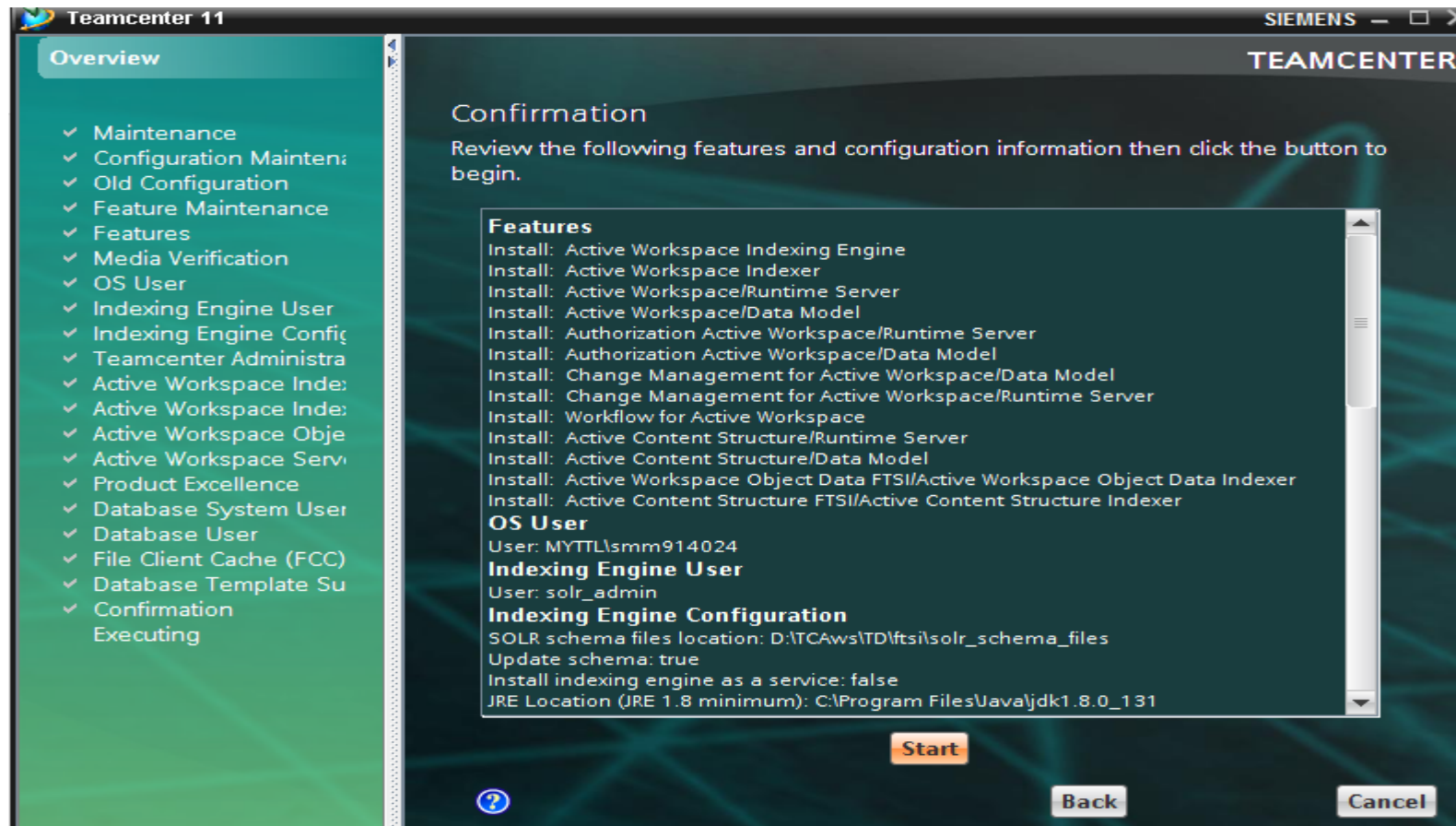
Install Server Extensions features

18. In the Database Template Summary panel, review the list the templates to be applied to the Teamcenter database. The templates listed vary depending on the features selected for installation.



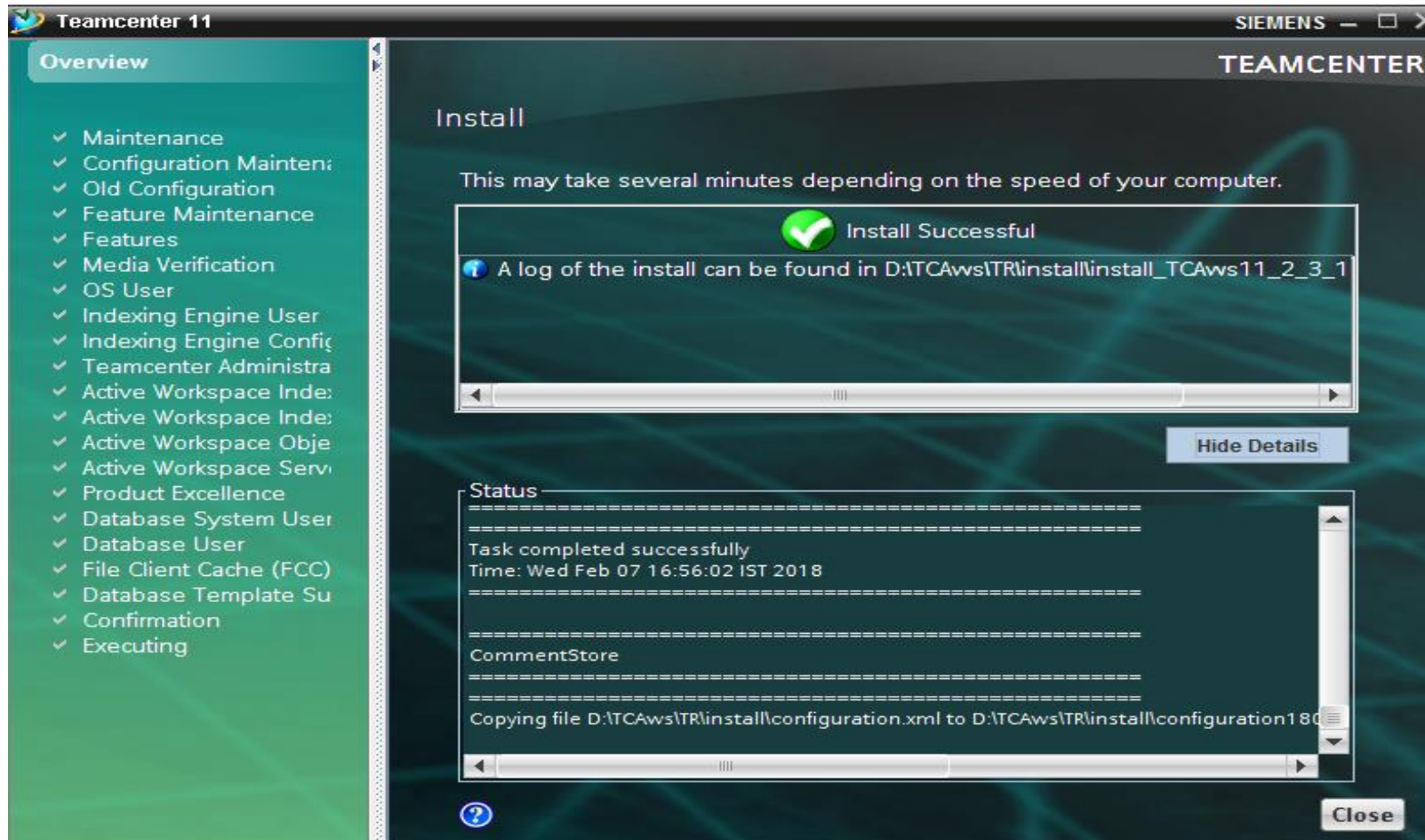
Install Server Extensions features

19. In the **Confirmation** panel, click **Start** to begin the installation

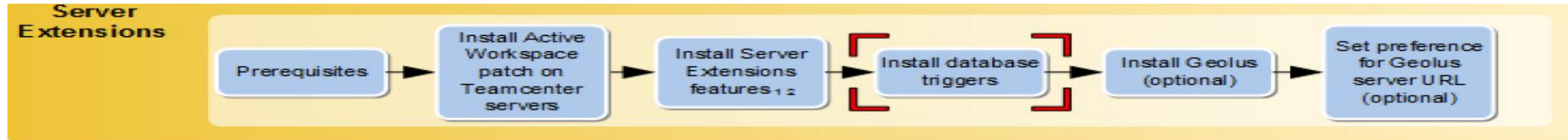


Install Server Extensions features

20. When installation is complete, close TEM



Install database triggers



The TcFTSIndexer requires database triggers to detect additions, modifications, and deletions to the database when performing run-time (synchronous) indexing.

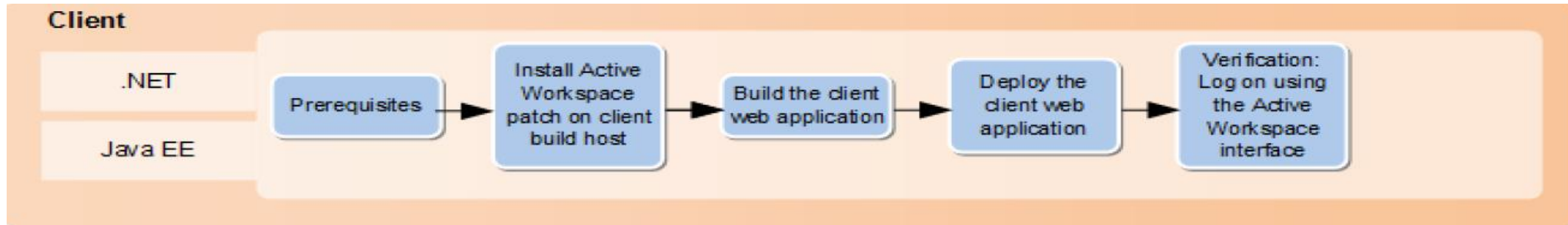
Under certain conditions, TEM may install the database triggers automatically during installation of Server Extensions. If TEM is not able to do this, TEM displays a TODO message notifying you to install database triggers manually.

Install database triggers

Install database triggers in Oracle

1. Grant **create trigger** privilege to the Oracle user that owns the Teamcenter database (typically **infodba**) by:
 - a. Open a command prompt.
 - b. Type:
sqlplus system/password
 - c. Type:
grant Create trigger to infodba identified by password;
 - d. Type:
exit
2. Create the trigger:
 - a. In the command prompt, type:
sqlplus infodba/password
 - b. Type:
@Teamcenter-installation-media\tc\install\sitecons\sitecons_install_triggers_oracle.sql

Client overview



Client functionality is provided by a user-facing web application that is built using TEM. Users connect to the client web application using a web browser. The web application can be either:

- A Java EE-based WAR file (**awc.war**) deployed on a Java EE web application server such as JBoss or WebSphere.
- A .NET-based web application deployed on Microsoft IIS.

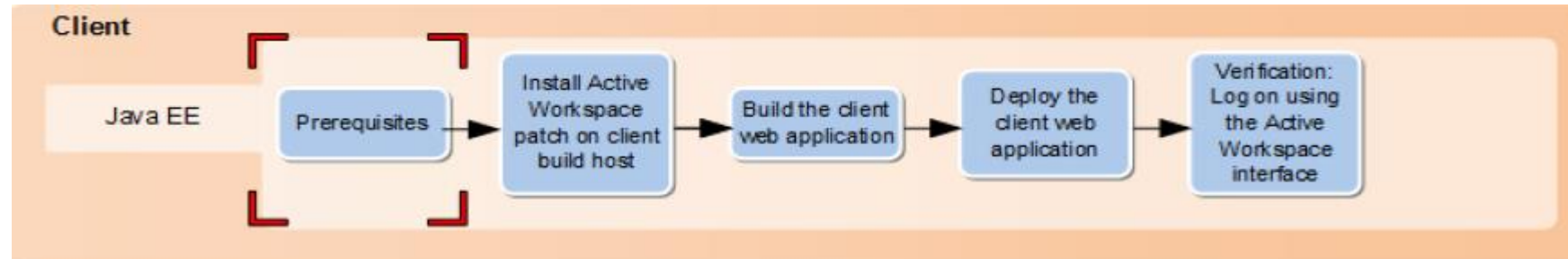
Note - The Active Workspace client web application must be installed (built) on a machine running a supported Windows operating system, but can be deployed on a machine running any operating system and application server on which the client web application is supported.

Java EE Client

Values required before beginning the Client taskflow		
Parameter	Where value is defined	Your value
Teamcenter four-tier URL	Deployment of Teamcenter four-tier web application	
JDK Home	JDK installation location on the machine used to build the awc.war file	
FSC Bootstrap URLs (http://host:4544)	Teamcenter installation	

Values to record during the Client taskflow execution		
Parameter	Where value is defined	Your value
Visualization Pool Assigner host and port	<i>Build the Java EE Client web application</i>	
Active Workspace Client URL	<i>Deploy the Java EE Client web application</i>	

Prerequisites for the Java EE Client



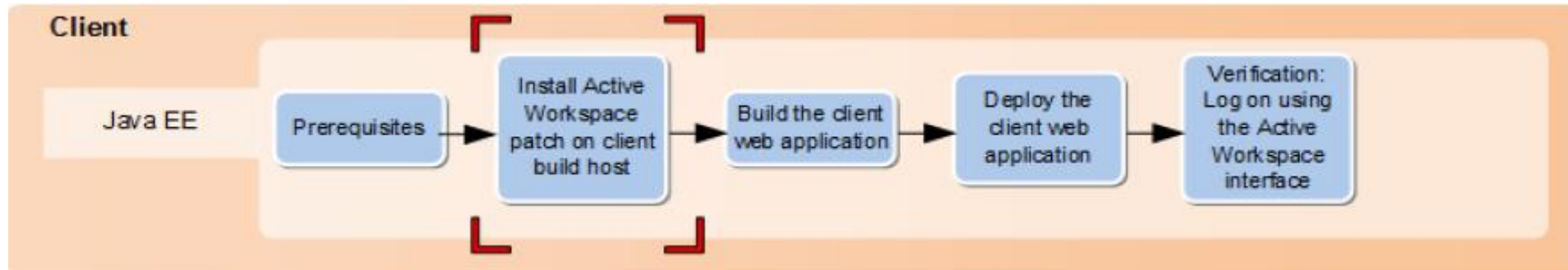
The machine on which you build the Active Workspace client web application must meet the following requirements. These requirements apply to the machine on which it is built, not to the machine on which it is deployed.

- A minimum of 1 GB of free memory to successfully generate the awc.war file. Siemens PLM Software recommends that the machine used have a minimum of 2 GB.
- Windows Server operating system.

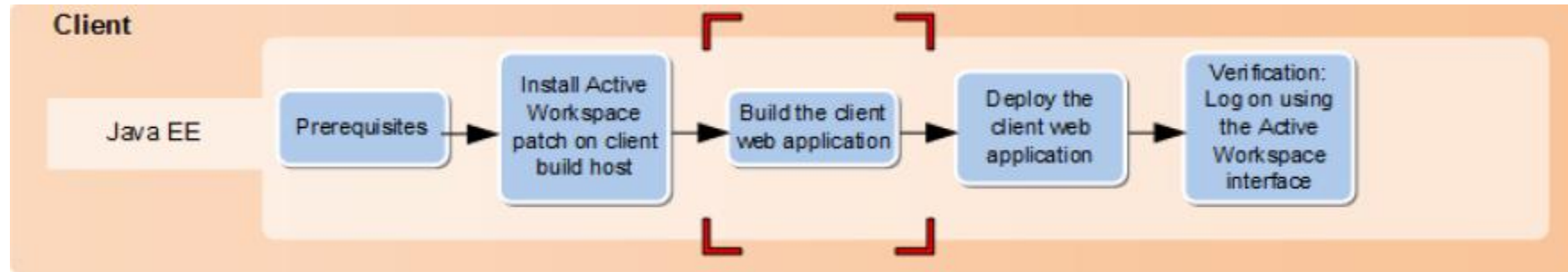
If you are running Teamcenter in a Linux environment, you can deploy the Java EE Active Workspace client web application on a Linux server, however, the **awc.war** file must be built on a Windows Server machine.

- JDK 64-bit is installed and the **JAVA_HOME** system environment variable must be set.

Install the Active Workspace patch on the Java EE Client build host



Build the Java EE Client web application



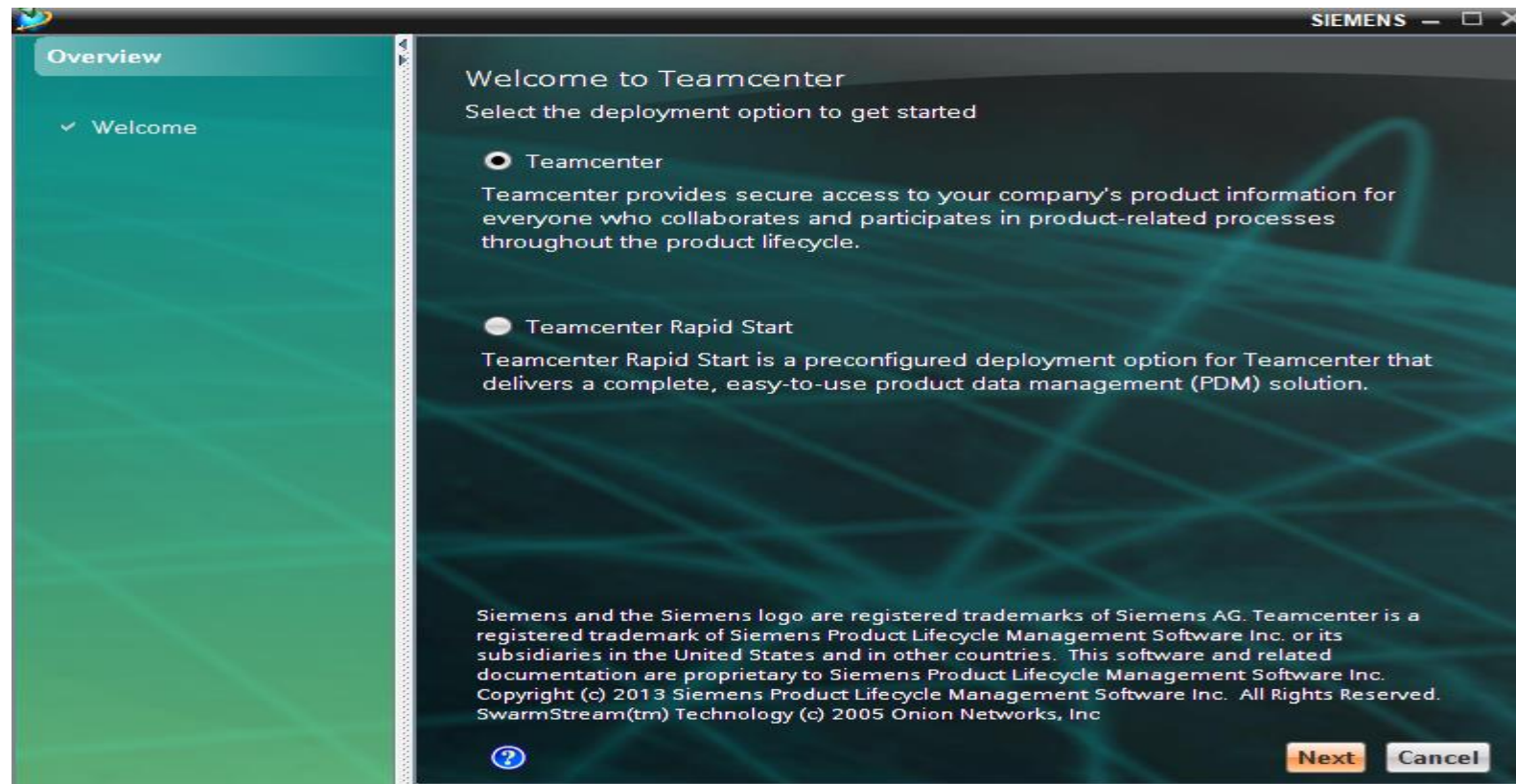
Make sure you have the following software kits available:

- Teamcenter major release
- Teamcenter minor release
- Active Workspace 4.2

Sanjoy_backup > backup > TC_12_2					Search TC_12_2	
	Name	Date modified	Type	Size		
	Tc12.0.0.0_ActiveWorkspace4.2.1_wntx64	27-01-2021 14:26	File folder			
	Tc12.0.0.0_wntx64	23-08-2021 17:31	File folder			
	Tc12.2.0_patch_7_wntx64	27-01-2021 14:32	File folder			

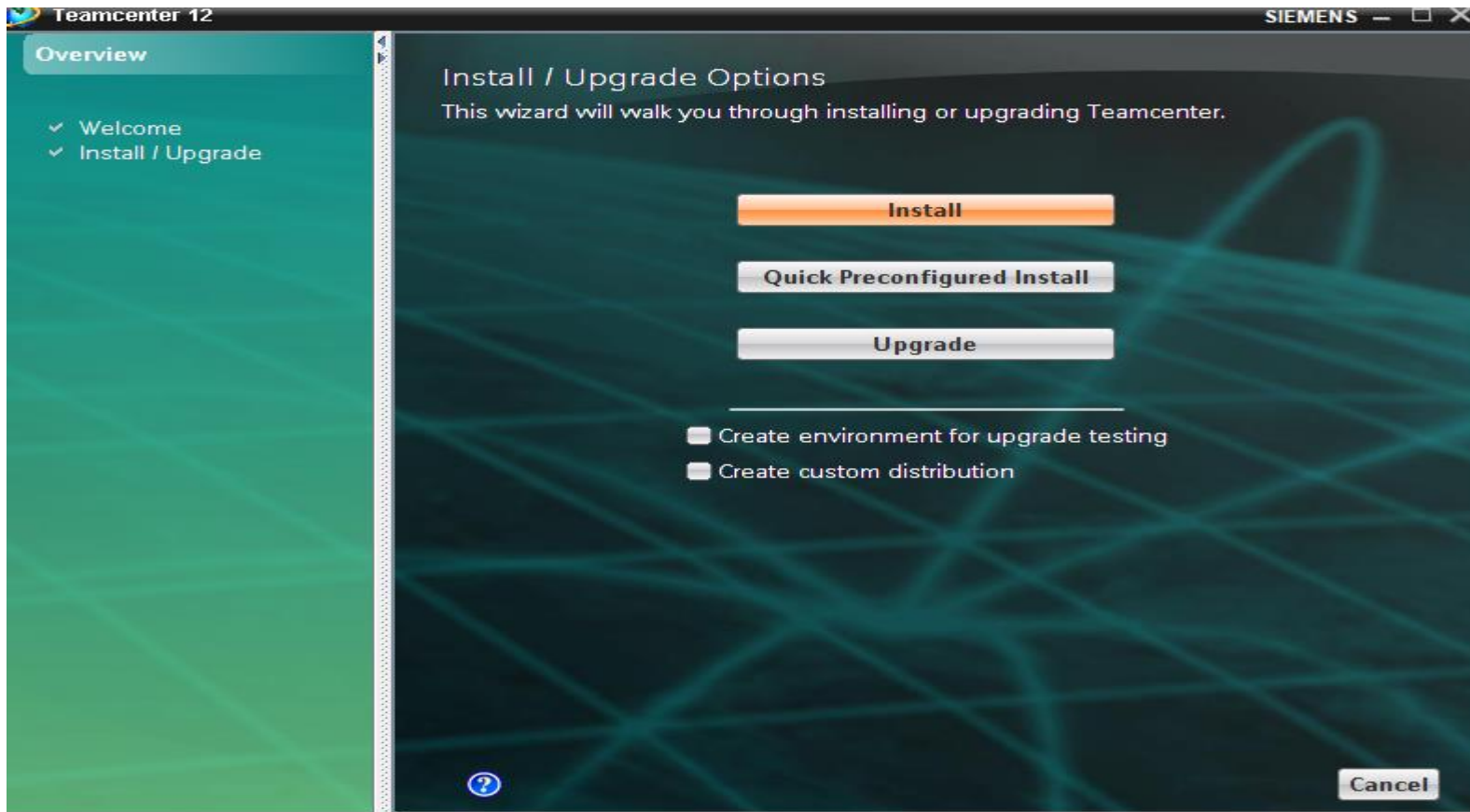
Build the Java EE Client web application

1. Launch TEM from the Teamcenter



Build the Java EE Client web application

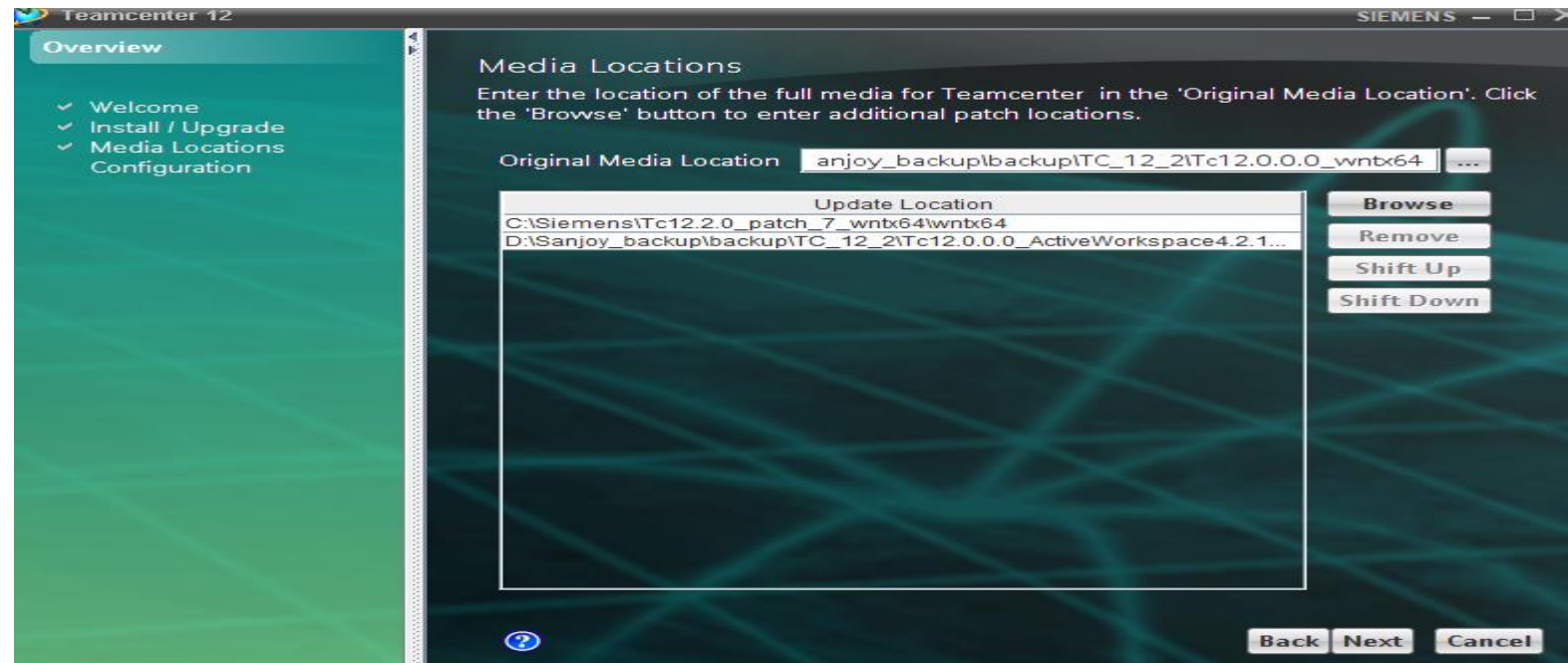
2. Select Install



Build the Java EE Client web application

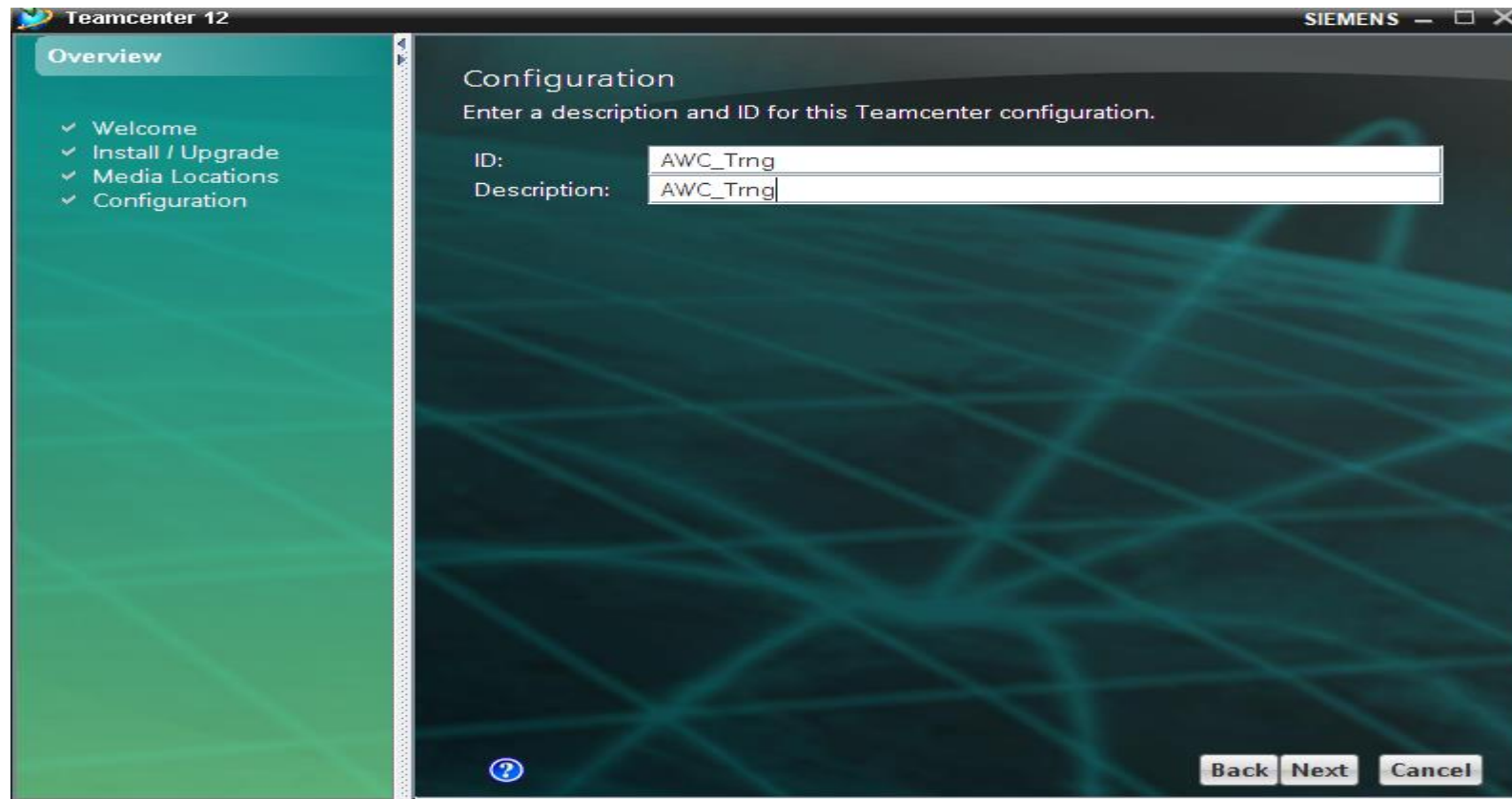
3. In the **Media Locations** panel, enter paths to the software kits.
 - In the **Original Media Location** box, enter the path to the software kit for the major version of Teamcenter or Rapid Start.
 - In the **Update Location** box, enter the path to Active Workspace 4.2 software kit.

The path to the Teamcenter or Rapid Start minor version software kit is already included in the list of update locations.



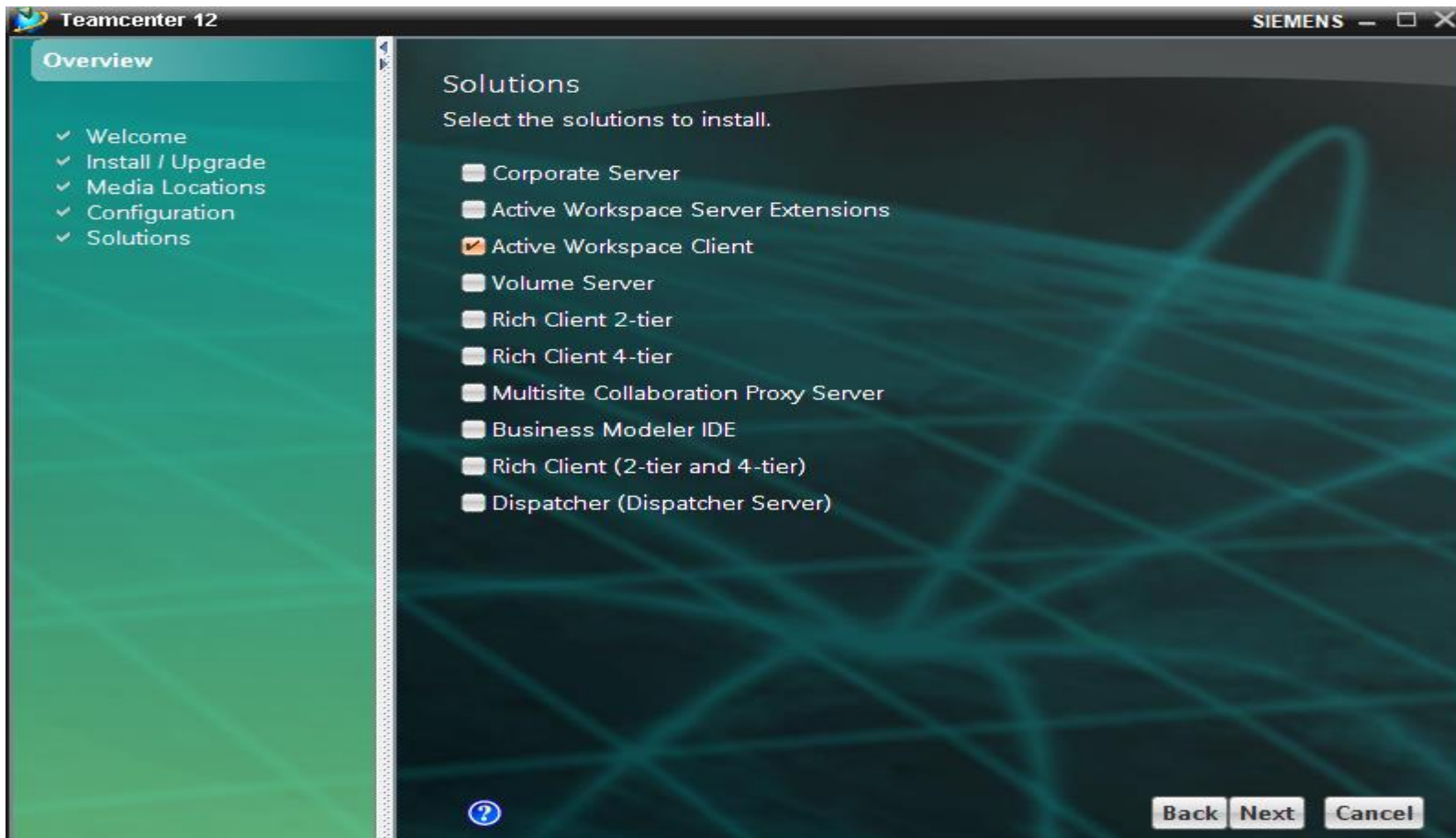
Build the Java EE Client web application

4. Enter Teamcenter configuration ID



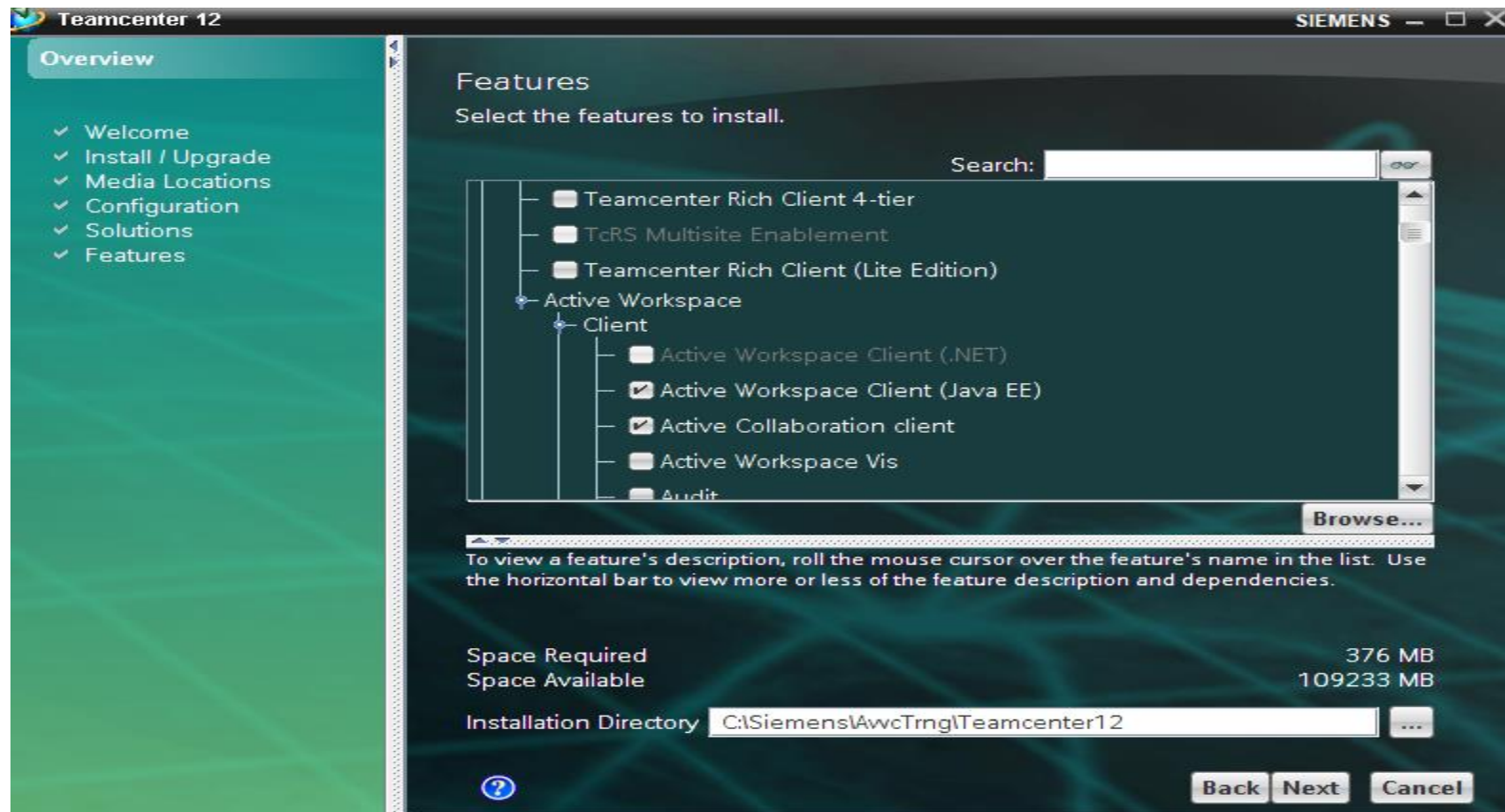
Build the Java EE Client web application

5. Select Active Workspace Client



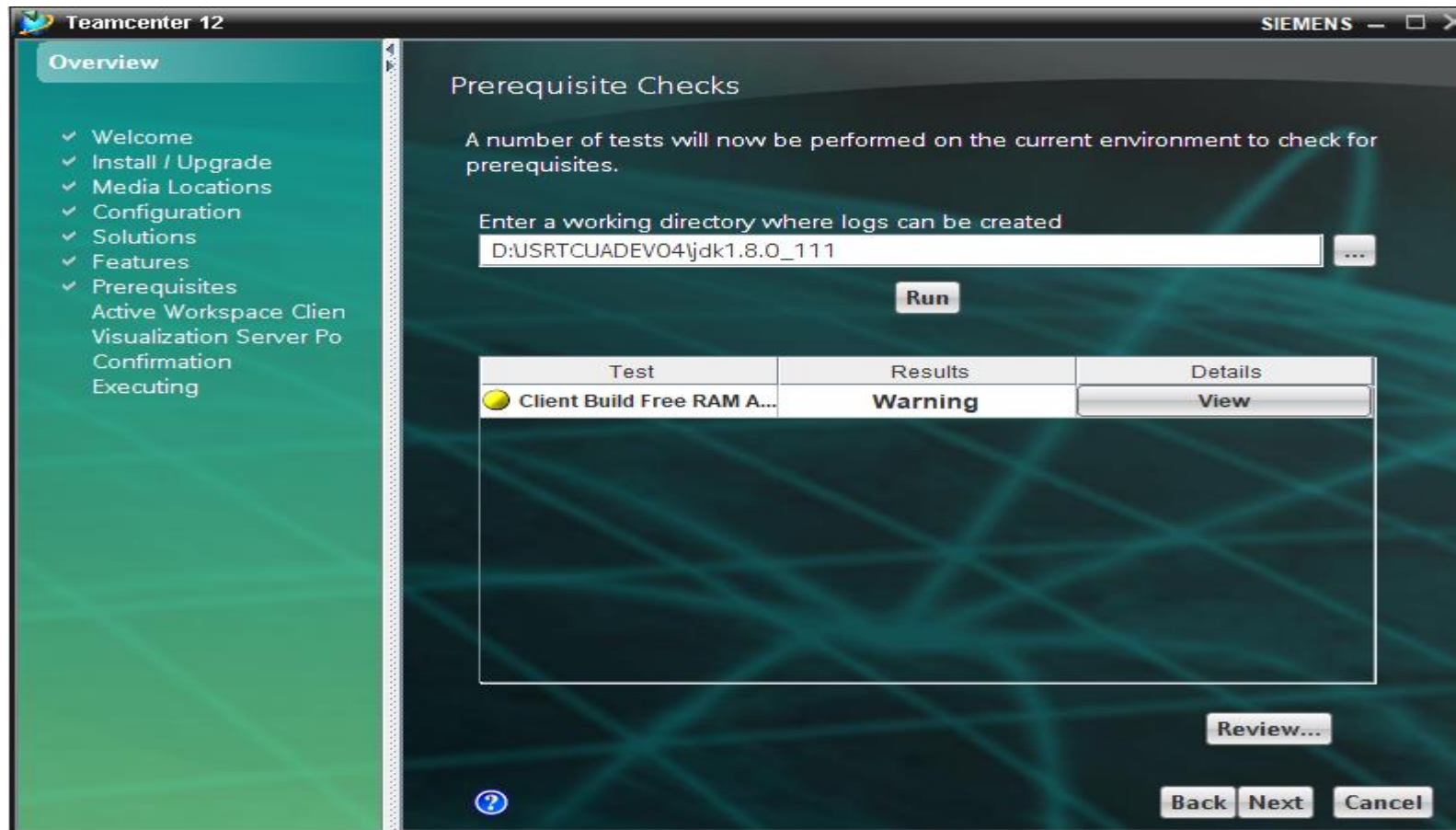
Build the Java EE Client web application

6. Select the Active Workspace Client features that you wish to install



Build the Java EE Client web application

7. Enter the JDK Path and click Run. This will check whether your current environment is compatible



Build the Java EE Client web application

8. Enter the Teamcenter 4 –tier Url and Bootstrap URLs

Teamcenter 12 SIEMENS

Overview

- ✓ Welcome
- ✓ Install / Upgrade
- ✓ Media Locations
- ✓ Configuration
- ✓ Solutions
- ✓ Features
- ✓ Prerequisites
- ✓ Active Workspace Client
- Administrative User
- Confirmation
- Executing

Active Workspace Client Settings

Specify Active Workspace client installation settings below.

Teamcenter 4-tier URL:

☒ Use as Bootstrap URLs

Bootstrap URLs:

Bootstrap Client IP:

☐ Use Assigned FSC URLs

Assigned FSC URLs:

Enable TcSS Support

☒ Enable TcSS Support

TcSS Application ID:

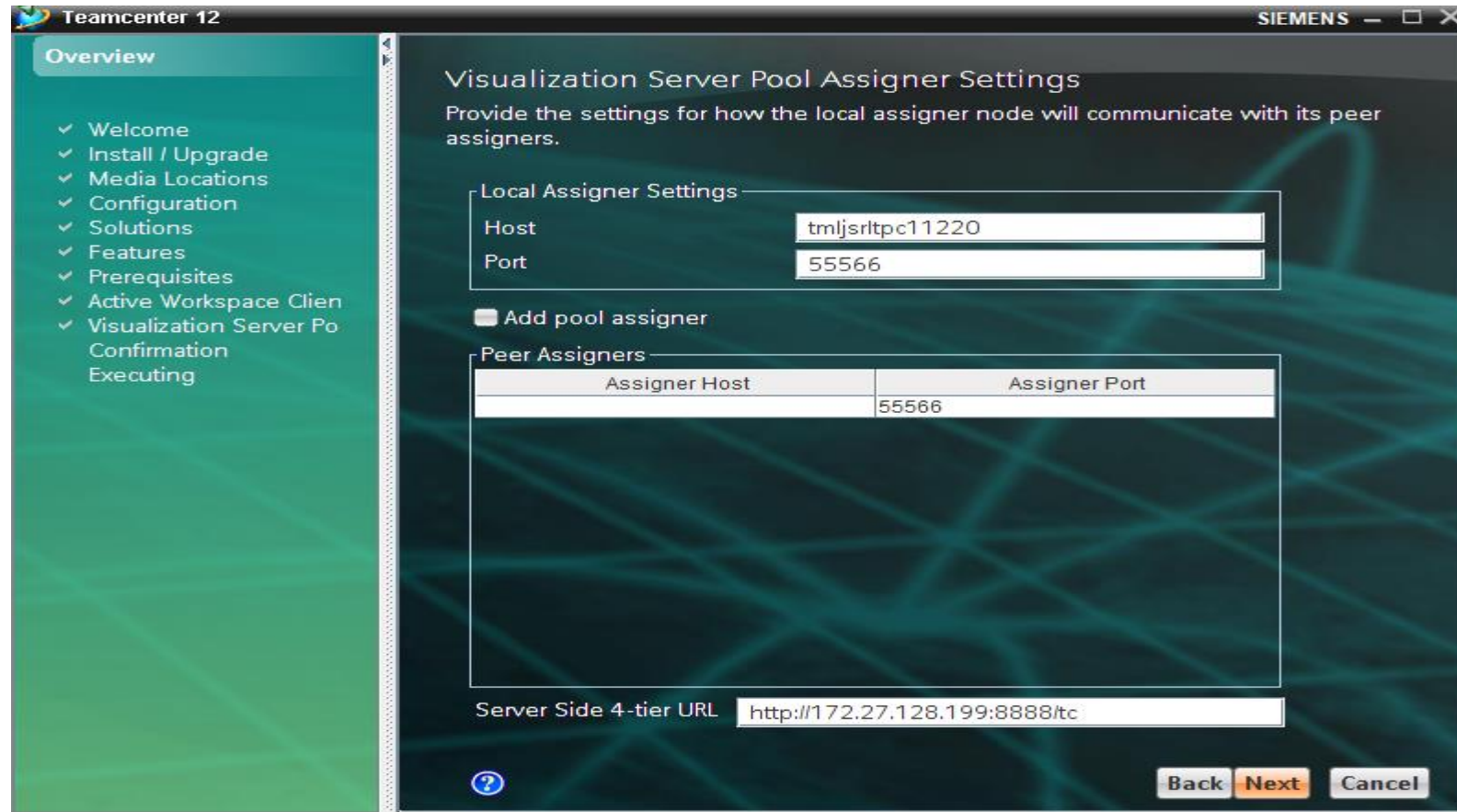
TcSS Login URL:

Advanced...

Back **Next** **Cancel**

Build the Java EE Client web application

9. Enter the details of Visualization Server Pool Assigner details



Teamcenter 12 SIEMENS

Overview

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- ✓ Active Workspace Client
- ✓ Visualization Server Pool Assigner Settings
- ✓ Confirmation
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Visualization Server Pool Assigner Settings

Provide the settings for how the local assigner node will communicate with its peer assigners.

Local Assigner Settings

Host:

Port:

☐ Add pool assigner

Peer Assigners

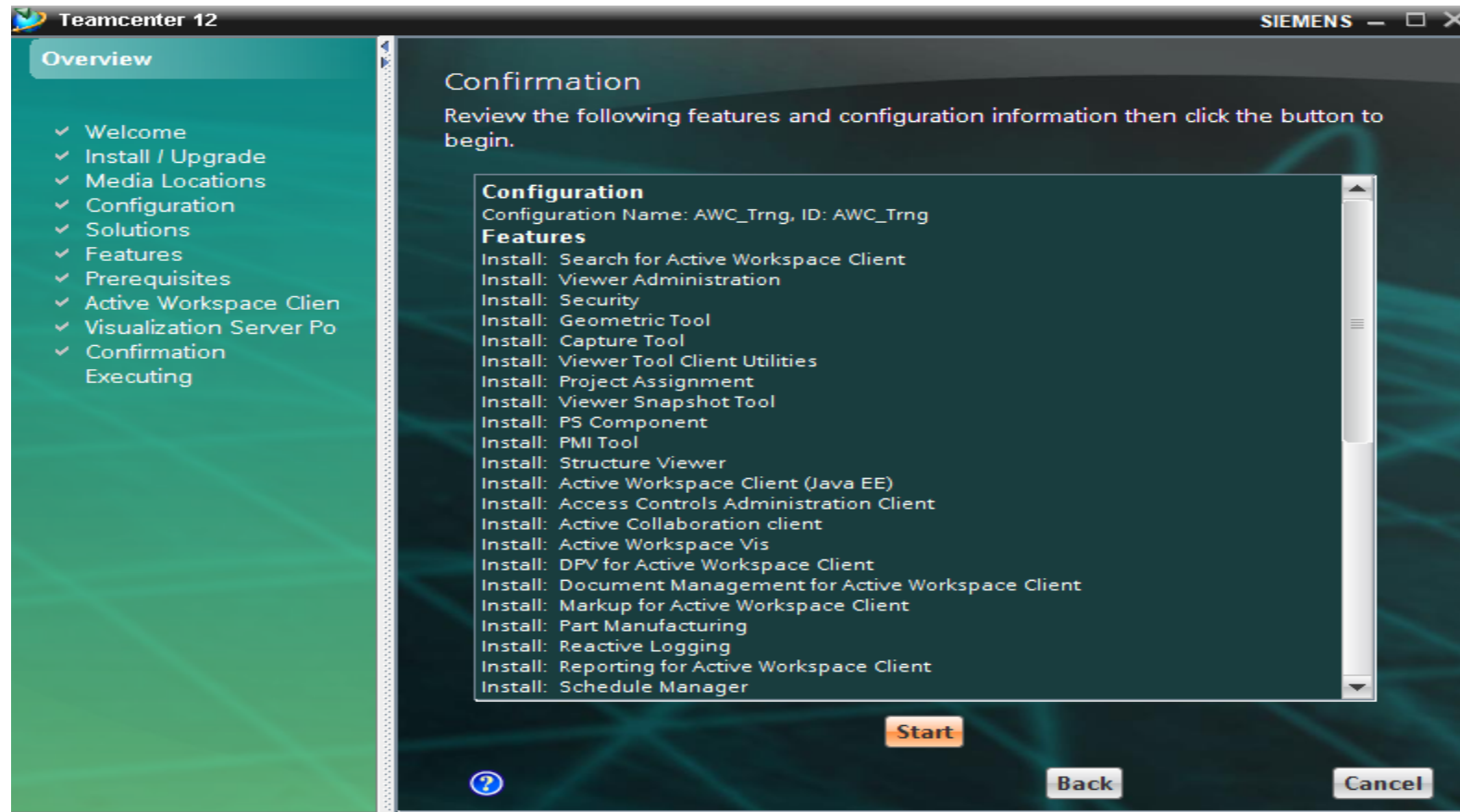
Assigner Host	Assigner Port
	55566

Server Side 4-tier URL:

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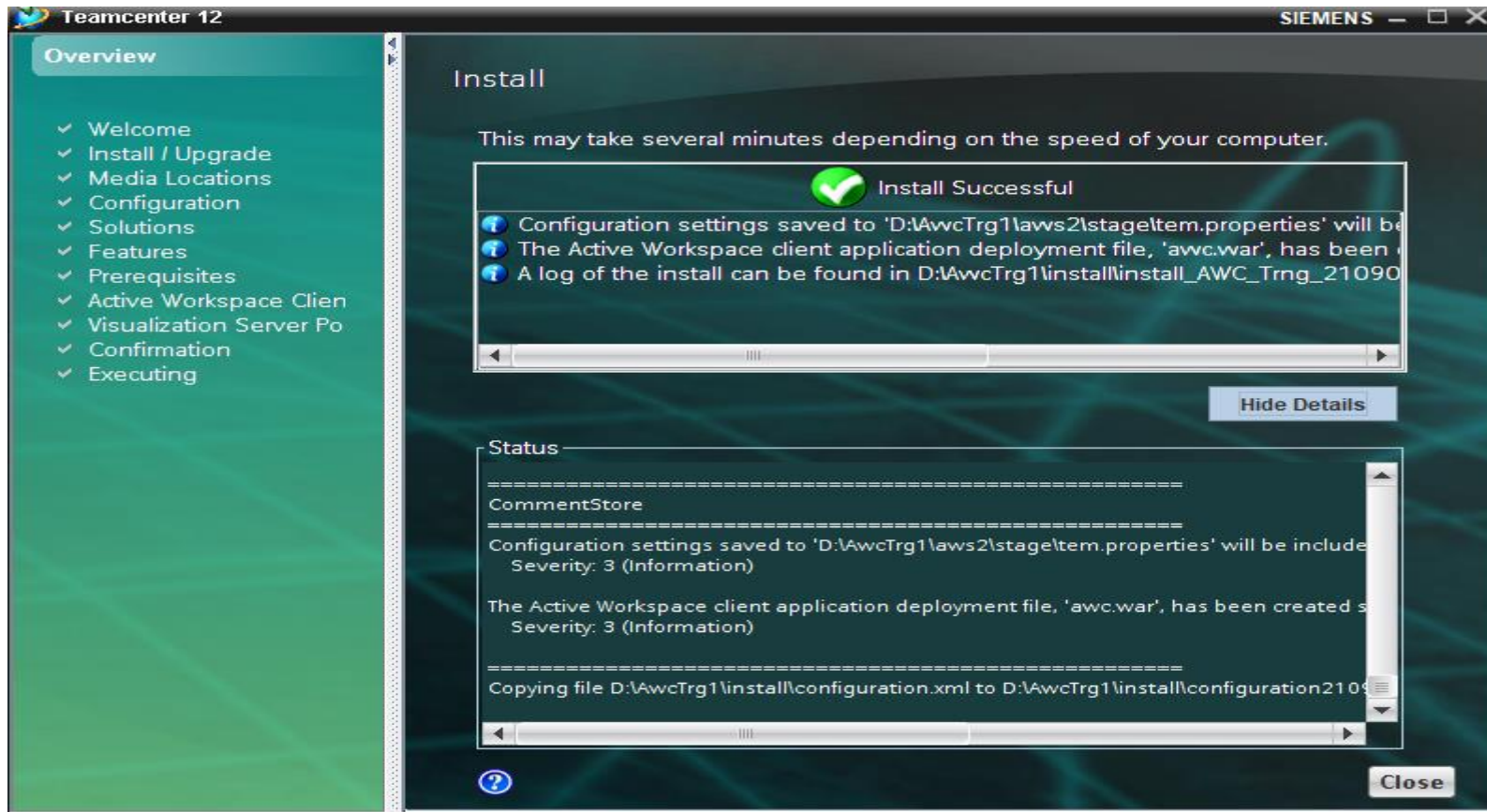
Build the Java EE Client web application

10. In the confirmation panel view the features that are going to be installed and select Start button to proceed.

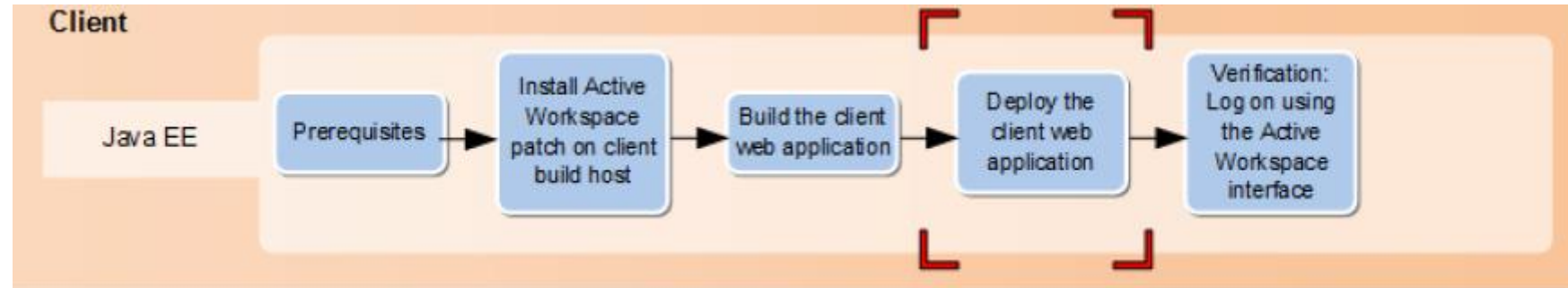


Build the Java EE Client web application

11. Once the installation completed close the TEM window



Deploy the Java EE Client web application

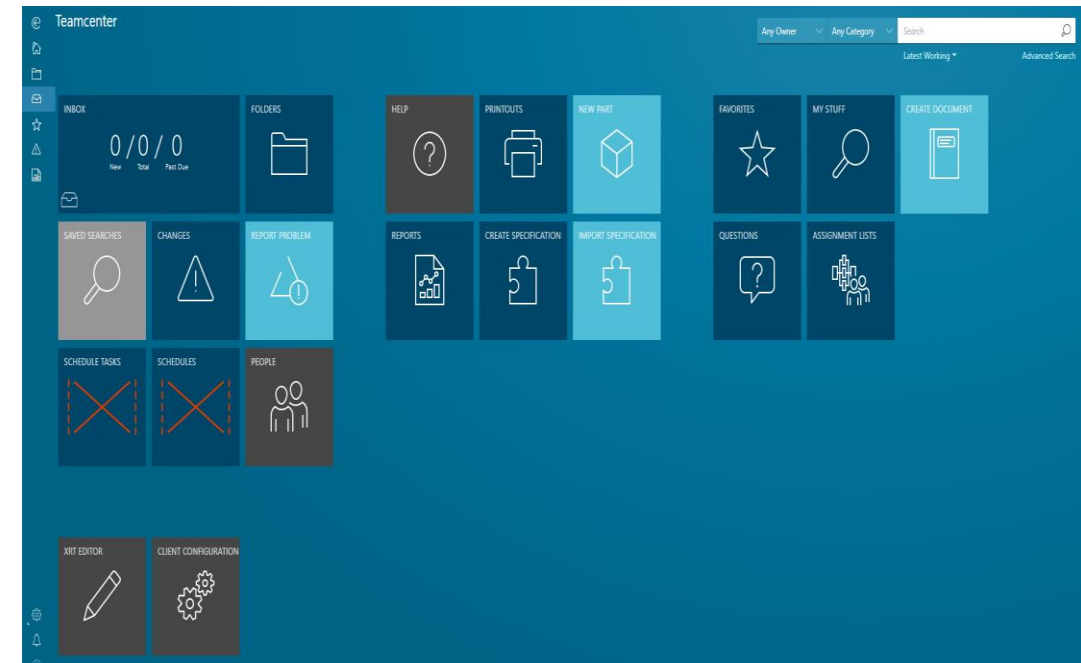
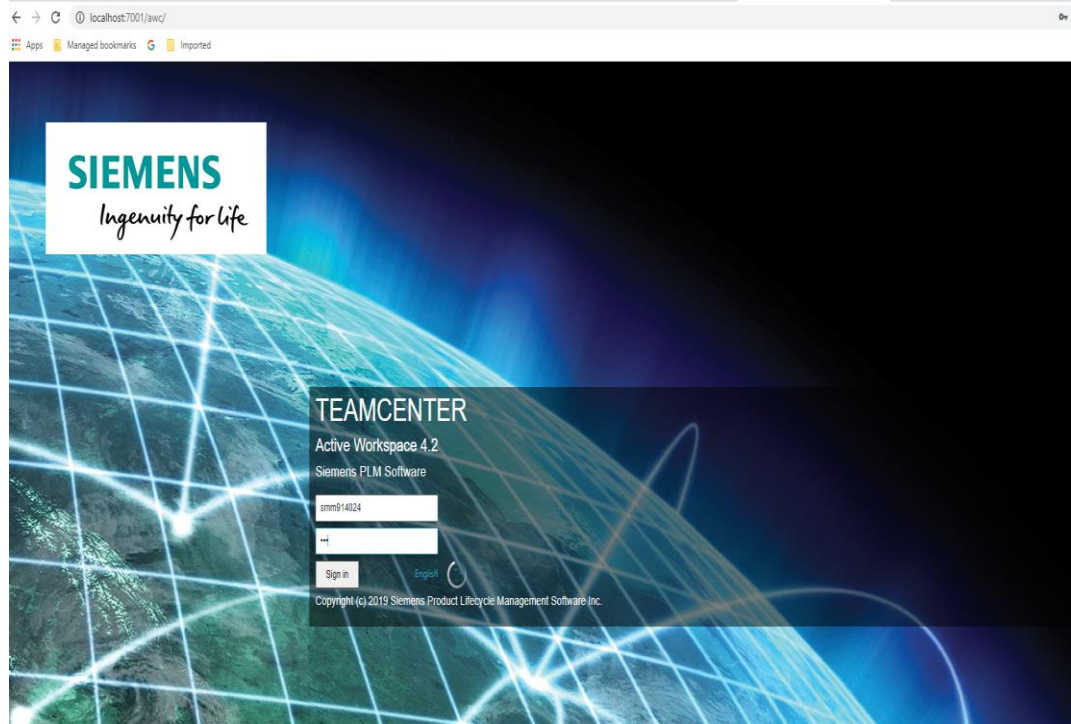
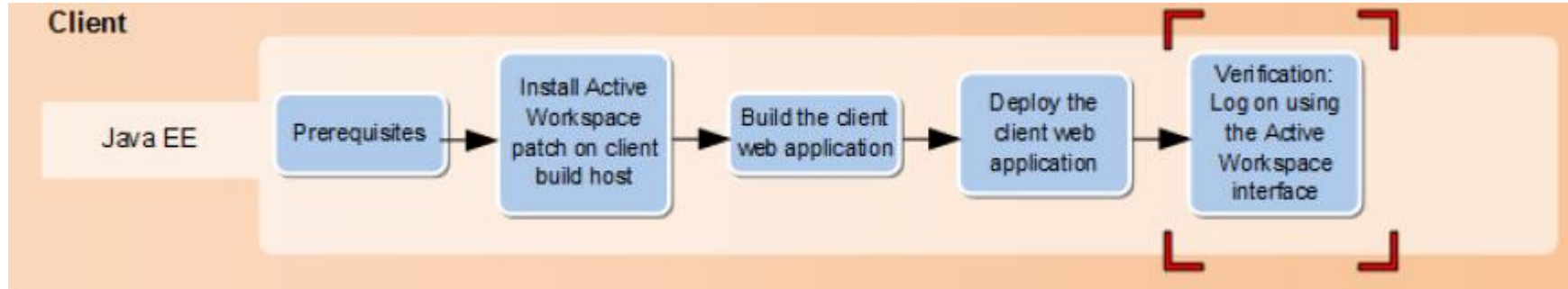


The Java EE Client WAR file (**awc.war**) is created in the `TC_ROOT\aws2\stage\out\` directory on the machine used to build it.

aws2 > stage > out				
Name	Date modified	Type	Size	
activeworkspace	02-09-2021 00:35	File folder		
awc.war	02-09-2021 00:37	WAR File	51,302 KB	
bundleData_tem.json	02-09-2021 00:36	JSON File	630 KB	

Deploy this awc.war file in any Java EE Web Application server e.g Tomcat or Jboss or WebSphere

Sign in using the Active Workspace interface



The Active Workspace development environment consists of three important directories. Not all directories present in the environment are shown.

1. stage

This is the base directory for Active Workspace customization. It is located in **TC_ROOT\aws2**. This is where the environment initialization script (**initEnv**) is located.

2. components

This directory contains the OOTB source for Active Workspace, and is required when building the web application. Optional components, like **darsi** and **gateway** for example, may also be found here.

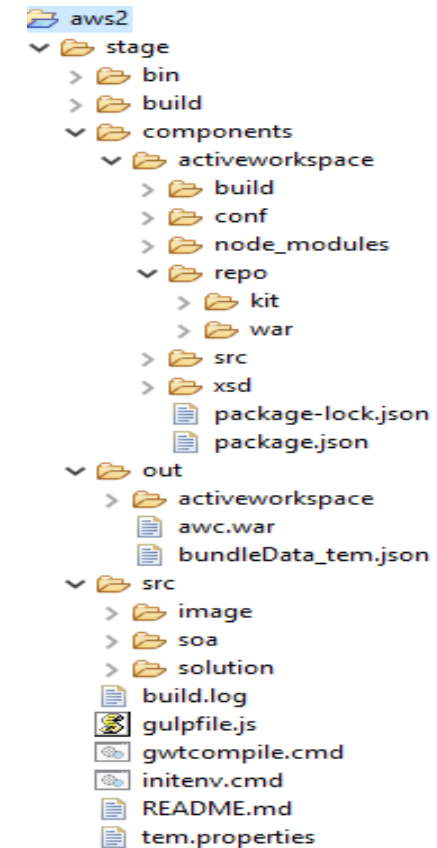
You can use the declarative source definitions in the **activeworkspace\repo\kit** directory for reference; all of the installed OOTB kits are located here in their respective directories.

3. out

This is where the build file can be found for deployment: **awc.war** for J2EE, and **awc.zip** for IIS. If the Gateway Client and the UI Builder are installed, you may find their working directories here as well.

4. src

The files and directories located here are for your customizations. They are pre-populated with actual source code and resources that are built along with the contents of the components directory.



The Active Workspace development scripts

To create custom Active Workspace components, you must work within an environment configured for Active Workspace development. Various scripts have been provided to assist in the development of component modules. Unless otherwise stated, all scripts must be run from the **stage** directory.

initEnv

Open a command prompt, and then run this script to configure the environment variables necessary for Active Workspace development. This only needs to be run once - when first opening a command prompt, before doing any other development work.

gwtCompile

This script compiles and builds the **awc** web application. By default, this compiles and builds the Active Workspace web application file: **awc.war** for J2EE or **awc.zip** for IIS.

This must be run from the **stage** directory.

generateModule

This script assists you in creating all the boilerplate files and directories for individual components. This script can be run from anywhere, it will place its artifacts in their proper locations. The **generateModule** script does not require any arguments. If you do not provide any, it will prompt you to enter the information it requires.

THANK YOU